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Return Extrapolation and Volatility Expectations

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Abstract

This article provides the first comprehensive evidence that the return extrapolation behavior of investors leads to biases in the expectations of volatility. Lower past returns are associated with higher expectations of volatility when using the physical, risk-neutral, and survey measures to estimate volatility expectations. Consistent with the return extrapolation framework, recent past returns have a larger impact than distant past returns on volatility expectations. Biases in volatility expectations are i) distinct from extrapolating past realized volatility, ii) asymmetrically induced by recent past negative returns, and iii) lead investors to pay more to insure against the perceived higher expected volatility.

I. Introduction

Classical asset pricing theories are based on representative agents with rational expectations. However, heterogeneity in information and investor beliefs, including biased beliefs, is an important feature of financial markets.[1] For example, Greenwood and Shleifer (2014), Kuchler and Zafar (2019), and Da, Huang, and Jin (2021) all provide evidence of biased beliefs due to extrapolative expectations (hereafter extrapolative beliefs). Barberis, Greenwood, Jin, and Shleifer (2015) develop a consumption-based model, which allows for extrapolative expectations by some investors and is consistent with well-known features of stock returns. In

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[1]Deviations from rational expectations and evidence of heterogeneity in investor expectations about future market returns can be seen in the Gallup investor survey, the American Association of Individual Investors survey, and the Investor Intelligence survey of investment newsletters.

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addition, Engelberg, McLean, and Pontiff (2018) show that biased expectation is one of the drivers of return anomalies.

While the above studies address the implications of extrapolative beliefs for expected returns, surveys suggest that investors also expect higher volatilities following low returns (Amromin and Sharpe (2014) and Kaplanski, Levy, Veld, and Veld-Merkoulova (2016)). Motivated by the survey evidence, Atmaz (2022) develops a dynamic equilibrium model wherein return extrapolation leads to biased variance expectations. This article provides the first comprehensive empirical evidence about the mechanism of extrapolative beliefs that leads to biased volatility expectations.[2]

Our tests include a comprehensive set of volatility expectations ($ExpVol_T$) proxies that span across physical expectations, risk-neutral expectations, and surveys-based expectations. We construct volatility expectations under the physical measure with i) a GARCH(1,1) model, ii) the Drechsler and Yaron (2011) model, and iii) the heterogeneous autoregressive realized volatility model of Corsi (2009). We use the CBOE VIX index and the Black–Sholes implied volatility of S&P 500 index options to proxy for the risk-neutral expectations. Finally, we construct survey-based expectations proxies from the Graham and Harvey CFO surveys and the Shiller stock market crash confidence surveys. This comprehensive set of volatility expectation measures provides consistent results about the return extrapolation effect on volatility expectations.

We adopt the MIDAS model of Ghysels, Santa-Clara, and Valkanov (2006) to examine whether return extrapolation affects volatility expectations. The results are as follows: Lower past returns are associated with higher expectations of volatility under the physical, risk-neutral, and survey measures. Consistent with the extrapolation framework, the recent past returns have a larger impact than the distant past returns on the volatility expectations. These findings differ from the well-known leverage effect in the volatility time-series modeling as the return extrapolation effect hinges on the differential contributions from recent versus distant past returns on future volatility expectations. Furthermore, the return extrapolation effect on volatility expectations is asymmetric. Past negative returns primarily drive our findings, while past positive returns have a much weaker impact on volatility expectations. These results are consistent across all key variables of interest, including i) the volatility expectations, ii) the variance risk premium, iii) the increase in volatility expectation, and iv) the unexpected shocks to volatility expectations.

Further, the impact of extrapolative beliefs on volatility is more prominent during periods of investor pessimism, when the market returns are lower, suggesting that it is investors' pessimistic sentiment that drives the asymmetric extrapolative beliefs in volatility expectations. Trading activity in out-of-the-money (OTM) put options is significantly larger than in OTM call options in months following negative market returns, suggesting that investors are impacted more by losses than gains, as implied by the prospect theory of Kahneman and Tversky (1979).

[2]Prior papers such as Jackwerth and Vilkov (2019) relate past returns to volatility expectations, while Amromin and Sharpe (2014) and Kaplanski et al. (2016) use surveys of expected volatilities. However, these papers do not link the biased volatility expectations to biased beliefs due to return extrapolation.

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Moreover, return extrapolation has asset pricing implications. Investor hedging demands, proxied by the change in the implied volatilities of OTM put options, increase more following negative returns.

In addition, negatively skewed past returns are related to return extrapolation. This is consistent with the model prediction of Atmaz (2022) that a higher degree of return extrapolation is associated with a larger magnitude of fundamental variance elasticity and, thus, with more negatively skewed returns. Moreover, a higher degree of extrapolation is associated with a larger volatility expectation bias. The sentiment mechanism (Proposition 3) of Atmaz (2022) suggests that extrapolating lower (higher) past returns simultaneously results in lower (higher) return expectations and higher (lower) volatility expectations. This mechanism leads to a negative relation between return expectations and volatility expectations, which is what we find.[3]

The effects of return extrapolation on volatility expectations are as follows: First, extrapolating past lower returns results in a higher variance risk premium, which is the difference between volatility expectations under the risk-neutral and physical measures. Also, the implied volatilities of the out-of-money put options increase. Hence, after extrapolating past lower returns, investors not only have higher volatility expectations but are also willing to pay a higher price to insure against a higher perceived future variance. Second, return extrapolation impacts the change in volatility expectations. Specifically, past returns are inversely associated with the change in volatility expectations (i.e., lower past returns lead to an increase in future expected volatility). Third, the unexpected shocks to the expectations of volatility exhibit a similar return extrapolation structure after controlling for the change in realized volatility.[4] Alternative explanations, such as the persistence of volatility or the slow-moving beliefs about volatility documented in Lochstoer and Muir (2023), cannot capture the impact of return extrapolation on volatility expectations. Collectively, these results suggest that extrapolating past lower returns induces an upward bias in volatility expectations, confirming the Atmaz (2022) model prediction that return expectations formed with extrapolative beliefs lead to biased volatility expectations.

The return extrapolation effect on volatility expectations exists at the stock level as well. Investors extrapolate a stock's past returns when forming expectations about its future return volatility. Similar to the index-level results, lower past returns are associated with higher volatility expectations, larger variance risk premiums, increases in volatility expectation, and more positive unexpected shocks to volatility expectations. All these effects are asymmetric and mainly driven by past negative stock returns. The results are stronger for stocks with higher market capitalization, lower market beta, more analysts following, and better stock liquidity. Overall, these findings show that the return extrapolation effect on volatility expectations is associated with a more transparent information environment and better liquidity. These findings echo the results of Da et al. (2021) that the extrapolative bias in expected returns is stronger among stocks with better visibility.

[3]Atmaz (2022) and Barberis et al. (2015) define the sentiment as the weighted average of the past returns.

[4]The higher (lower) unexpected shocks capture the over (under) estimation bias in the volatility expectation.

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Our study contributes to the literature in two ways. First, we add to the burgeoning literature on extrapolative beliefs in asset pricing (Greenwood and Shleifer (2014), Barberis et al. (2015), Cassella and Gulen (2018), and Da et al. (2021)).[5] While these studies focus on the impact of return extrapolation on return expectations, we provide the first comprehensive examination of whether extrapolating past returns also induces biased beliefs about volatilities at both the market and the individual stock level. In this regard, our study provides empirical support for the model implications of Atmaz (2022), who explicitly incorporates the impact of return extrapolation on volatility expectation into options prices. Atmaz (2022) shows that implied volatilities are affected by the return extrapolation that induces biased beliefs about future volatilities.

Unlike papers that document a negative relation between past returns and expected volatility, we explicitly link this relation to return extrapolation. Our results also complement those of Amromin and Sharpe (2014) and Kaplanski et al. (2016) in that the return extrapolation effect induces biases in the volatility expectations, which, in turn, results in a negative risk–return trade-off ex ante. Moreover, our article is also related to Goyal and Saretto (2009) who argue that a large deviation between implied volatility and the long-run average historical volatility is a signal of mispricing based on the mean-reversion feature of realized volatilities. They attribute their findings to the overreaction of stock returns, as suggested by Barberis and Huang (2001). In contrast, our results and interpretations are rooted in the extrapolative belief framework (e.g., Barberis et al. (2015) and Atmaz (2022)) wherein investors focus more on *recent* past returns rather than distant past returns when forming expectations about future volatilities.

Second, our study provides new insights into the mechanism underlying extrapolative beliefs. We show that the impact of return extrapolation on volatility expectations is more pronounced following periods of low returns and is predominantly driven by past negative returns. This finding suggests that the mechanism for extrapolative belief in volatility is linked to investor pessimism. In contrast, the mechanism for extrapolative belief in return expectations is found to be associated with investors' optimistic sentiment. For example, Barberis et al. (2018) argue that extrapolating good news about fundamentals can trigger price bubbles. Our article complements Barberis et al. (2018) as our results highlight the differences in the return extrapolation-driven expectations of the first and second moments of returns.

II. Key Variables and Econometric Methodology

A. Key Variables

The key variables for the analysis are the same as those used in the examination of the return extrapolation effect: i) the expectations of volatility, $ExpVol_T$, ii) the variance risk premium, VRP_T, iii) the change in volatility expectations,

[5]Other studies on extrapolative beliefs include Alti and Tetlock (2014), Barberis et al. (2015), Barberis, Greenwood, Jin, and Shleifer (2018), Hirshleifer, Li, and Yu (2015), Frydman and Nave (2017), Glaeser and Nathanson (2017), Choi and Mertens (2019), Hartzmark, Hirshman, and Imas (2021), Atmaz (2022), Egan, MacKay, and Yang (2022), Jin and Sui (2022), Pénasse and Renneboog (2022), Beutel and Weber (2022), and Li and Liu (2023).

$\Delta ExpVol_T = ExpVol_T - ExpVol_{T-1}$, and iv) the unexpected volatility expectation shocks, ε_T.

1. Volatility Expectations

We construct several measures of volatility expectations. Unlike prior studies that rely on surveys for the expectation of returns, we include a comprehensive set of proxies for volatility expectations, including risk-neutral expectations, physical expectations, and survey expectations. The risk-neutral expectations of volatility are the CBOE (Chicago Board of Options Exchange) VIX index and the implied volatility based on the Black–Scholes model. The CBOE uses a model-free approach to compute VIX from the highly liquid S&P 500 index options. Using VIX to proxy for expected volatility avoids concerns that the implied volatility calculated from a specific model, such as Black–Scholes, may be biased due to the model specification. Since its inception in 1993, both practitioners and academics have viewed the VIX index as a benchmark measure for the market's expectation of 30-day volatility (Bollerslev et al. (2009)). Specifically, at the beginning of each month T, the VIX index (VIX_T) is used as a proxy for the market expectations of volatility over the following month. The Black–Scholes implied volatility, IV_T, is obtained from at-the-money S&P 500 index options observed at the beginning of the month T.

The three physical conditional volatility expectation measures are i) VOL_T^{GARCH}, where the month T conditional volatility expectation is forecasted based on a GARCH(1,1) model, ii) VOL_T^{DY}, where the month T conditional volatility expectation forecast uses the Drechsler and Yaron (2011) model, and iii) VOL_T^{Corsi}, where the month T conditional volatility expectation forecast uses the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) of Corsi (2009). It is important to note that these three physical measures are out-of-sample forecasts that use data in a 60-month rolling window with the last observation at the end of month $T-1$. In the Appendix, we provide details about the construction of these conditional physical expectation measures.

Finally, we construct survey-based volatility expectations, using i) Graham and Harvey's CFO surveys, and ii) Shiller's stock market crash confidence survey. While the survey-based proxies offer complementary evidence to the risk-neutral and physical volatility expectation measures, they have their own limitations (Brunnermeier, Farhi, Koijen, Krishnamurthy, Ludvigson, Lustig, Nagel, and Piazzesi (2021)). For example, survey respondents may not be marginal investors. In addition, there are two specific issues with the survey data in our setting.

First, these surveys do not directly ask about volatility expectations. Instead, Graham and Harvey's CFO survey asks correspondents about the 10th and 90th percentiles for the expected return distribution. Shiller's stock market crash survey asks individual and institutional investors separately about the probability that the stock market will crash (like that of Oct. 28, 1929 or Oct.19, 1987) in the next 6-month period and reports the percentage of respondents who think that the probability is strictly less than 10%. We take the difference between the 90th and 10th percentile returns distribution for the CFO survey-based volatility expectation measure following Lochstoer and Muir (2023) and use the negative of the Shiller Stock Market Crash Index as the volatility expectation measure following Nagel and Xu (2023). $VOL_T^{CFOSurvey}$ is the volatility expectation from the CFO

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surveys, and $VOL_T^{Shiller_Ind}$ and $VOL_T^{Shiller_Ins}$ are the volatility expectations from Shiller's stock market crash survey of individual and institutional investors, respectively.

Second, the participants respond to surveys over a period of time.[6] This window creates further challenges to properly identify the information set, based on which the investors are forming the expectations. In contrast, the market price-based risk-neutral and physical volatility expectations allow for the precise identification of the timing of the investor information set. Also, these survey-based proxies inevitably have more measurement noise than the risk-neutral and physical volatility expectations. Nevertheless, we include the survey-based proxies into our analysis to potentially capture the subjective expectations of investors.

2. Variance Risk Premium

To gain additional insights into the impact of return extrapolation on volatility expectations, we examine how the return extrapolation influences the variance risk premium (VRP). The VRP is the difference between the variance expectations under the risk-neutral $\mathbb{E}_T^Q(Var)$ and physical $\mathbb{E}_T^P(Var)$ measure. Carr and Wu (2009) argue that VRP is equivalent to the ex ante unit cost of hedging the variance risk.[7] If return extrapolation affects investor volatility expectations, the impact should also be reflected in the VRP since risk-averse investors would be willing to pay more (higher VRP) to hedge against the higher perceived risks following lower returns.[8]

At the beginning of each month T, VRP_T is the difference between VIX_T^2 (i.e., the variance expectation under the risk-neutral $\mathbb{E}_T^Q$ measure) and a physical expectation measure $\mathbb{E}_T^P(Var)$. As discussed, we have several VRP_T measures constructed using different physical measures, $\mathbb{E}_T^P(Var)$. VRP_T^{GARCH} utilizes the physical measure from the GARCH(1,1) model, VRP_T^{DY} uses the model of Drechsler and Yaron (2011), VRP_T^{Corsi} uses the HAR-RV model of Corsi (2009), and VRP_T^{BTZ} uses the realized variance in the prior month as the physical measure following Bollerslev, Tauchen, and Zhou (2009).

3. The Change in Volatility Expectations

Next, we investigate the return extrapolation effect on the change in volatility expectations. Atmaz (2022) shows that return extrapolation induces a bias in the expectation of variance such that lower (higher) past returns lead to higher (lower) variance expectations with an overestimation (underestimation) bias. Atmaz (2022) further demonstrates that unlike the rational variance expectation $E_T^R[Var]$

[6]The Graham and Harvey's CFO survey is conducted over a quarterly cycle, and the survey is open for a month (the second month of the quarter). Shiller's stock market survey is conducted monthly. However, the index only reports the 6-month moving average of the survey results and not the calendar month values.

[7]Carr and Wu (2009) show that the risk-neutral expectation of variance is essentially the variance swap rate. The difference between the variance swap rate and realized variance is the unit cost of insuring against volatility risk.

[8]We will also examine the impact of return extrapolation on the implied volatilities of out-of-money put options.

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(i.e., without extrapolative belief), the subjective variance expectation with the extrapolative belief $E_T^S[Var]$ is affected by the degree of extrapolation and sentiment elasticity.[9] However, quantifying the extrapolation bias is challenging since the benchmark rational expectation $E_T^R[Var]$ is not observed. To overcome this difficulty in measurement, we use the change in volatility expectations, $\Delta ExpVol_T$. Under the null of no bias in volatility expectations, a random walk model would suggest that $\Delta ExpVol_T = 0$ and $\Delta ExpVol_T$ should not be associated with extrapolating past returns. If return extrapolation induces persistent systematic biases in volatility expectations, we should expect that extrapolating past lower (higher) returns results in an overestimation (underestimation) bias in the $ExpVol_T$. This overestimation (underestimation) should be associated with increases (decreases) of $ExpVol_T$. Hence, $\Delta ExpVol_T$ should exhibit a significant return extrapolation structure and a negative relation with the past returns.

We compute $\Delta ExpVol_T$ measures that correspond to the different $ExpVol_T$ proxies defined earlier. ΔVIX_T and ΔIV_T are the monthly changes in the VIX index and the Black–Scholes implied volatility of the S&P 500 index options, respectively. ΔVOL_T^{GARCH}, ΔVOL_T^{DY}, *and* ΔVOL_T^{Corsi} are the monthly changes in conditional expected volatility based on the GARCH(1,1) model, the Drechsler and Yaron (2011) model, and the HAR-RV model of Corsi (2009), respectively. $\Delta VOL_T^{CFOSurvey}$, $\Delta VOL_T^{Shiller_Ind}$, and $\Delta VOL_T^{Shiller_Ins}$ are based on the surveys.

4. Unexpected Shocks to Volatility Expectations

Given that $\Delta ExpVol$ may have an AR structure, we also examine the effect of return extrapolation on unexpected shocks to volatility expectations. The unexpected volatility expectation shock is defined as the residual ε_T from the regression

$$\Delta ExpVol_T = \beta_0 + \sum_{i=1}^{n} \beta_{1,i} \Delta ExpVol_{T-i} + \sum_{i=1}^{n} \beta_{2,i} \Delta RVol_{T-i} + \varepsilon_T, \tag{1}$$

that regresses $\Delta ExpVol_T$ on the lagged $\Delta ExpVol_{T-i}$ and lagged $\Delta RVOL_{T-j}$, where the lag structure is determined by the Bayesian information criterion.[10] We use the residual ε_T as a measure of the unexpected innovation in $ExpVol_T$ to study the impact of biased beliefs on volatility expectations. A higher (lower) value of ε_T (i.e., a higher (lower) unexpected shock to $\Delta ExpVol_T$ after controlling for $\Delta ExpVol_{T-i}$ and $\Delta RVol_{T-i}$), reflects the over-reaction (under-reaction) of $ExpVol_T$. Investigating the relation between ε_T and past returns offers additional insights into the volatility expectation bias.

We estimate regression (1) with each of the $ExpVol_T$ proxies and derive the following measures of unexpected shocks. ε_T^{VIX} and ε_T^{IV} are the unexpected innovations in VIX_T and IV_T, respectively. ε_T^{GARCH}, ε_T^{DY}, and ε_T^{Corsi} are the unexpected innovations in VOL_T^{GARCH}, VOL_T^{DY}, and VOL_T^{Corsi}, respectively.

[9] See Propositions 3 and 4 in Atmaz (2022).

[10] The month T realized volatility, $RVol_T$, is the month-end realized volatility based on intraday 5-minute returns.

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B. Econometric Model

To study the return extrapolation effect, we first need to examine the relation between the dependent variables (i.e., $ExpVol_T$, VRP_T, $\Delta ExpVol_T$, and $\varepsilon_{1,T}$) and past returns, and then compare the contributions from the recent versus distant returns. While the dependent variables are all measured at a monthly frequency, the main independent variable (past returns) has to be measured at a daily frequency to distinguish between the contributions of recent and distant past returns. We thus adopt the mixed data sampling (MIDAS) regression framework of Ghysels et al. (2006) to accommodate different sampling frequencies between the dependent variables and independent variables. Moreover, MIDAS provides a parsimonious and flexible estimation for the weighting from each lagged daily return. Greenwood and Shliefer (2014) have argued that return extrapolation should lead to recent past returns having a larger impact than distant past returns. The MIDAS specification allows us to compare contributions from recent versus distant lagged returns.

We estimate four baseline MIDAS regressions, where the dependent variable, Y_T, is one of the volatility expectation measures ($ExpVol_T$), variance risk premium (VRP_T), change in volatility expectation measures ($\Delta ExpVol_T$), and measures of unexpected innovation in $ExpVol_T$ (ε_T). The MIDAS regression is as follows:

$$Y_T = \mu_1 + \phi_1 \sum_{t=1}^{22} b_1(t;\theta) Ret_{T-t} + \gamma_1 Y_{T-1} + X_{T-1} + \mu_{i,T}. \tag{2}$$

Equation (2) is motivated by the prediction equations in Greenwood and Shleifer (2014) and Da et al. (2021) but with volatility expectations instead of return expectations as the dependent variable.[11] We also include the lagged dependent variable Y_{T-1}, and the following additional control variables, X_{T-1}, i) Vol_{T-1}, lagged market volatility in the prior month $T-1$[12], ii) $Vol_{T-6,T-2}$, the average lagged monthly market volatility computed over months $T-6$ to $T-2$, and iii) VoV_{T-1}, the lagged volatility-of-volatility, defined as the standard deviation of the daily VIX index in month $T-1$. The inclusion of Vol_{T-1} and $Vol_{T-6,T-2}$ is motivated by the slow-moving beliefs about volatility documented in Lochstoer and Muir (2023), and VoV_{T-1} is motivated by Baltussen, van Bekkum, and van der Grient (2018) and Chen et al. (2022). The coefficient ϕ_1 is the scaling parameter that captures the overall relation between the high-frequency lagged daily excess returns and the low-frequency monthly dependent variable, Y_T. The coefficient $b_1(t;\theta)$ captures the weights on the lagged daily excess returns in predicting the dependent variables Y_T. Note that the capital letter T refers to the monthly frequency variables, and the lowercase t refers to the daily frequency variable. Figure 1 provides the details of the timing of the returns and volatility expectations. Following

[11] See Equation (3) in Greenwood and Shleifer (2014) and Equation (2) in Da et al. (2021).

[12] Since the CFO survey is a quarterly measure, and Shiller's stock market survey is based on a 6-month rolling window result, we do not include Vol_{T-1} as the control variable and calculate VoV_{T-1} over the previous quarter and 6 months respectively.

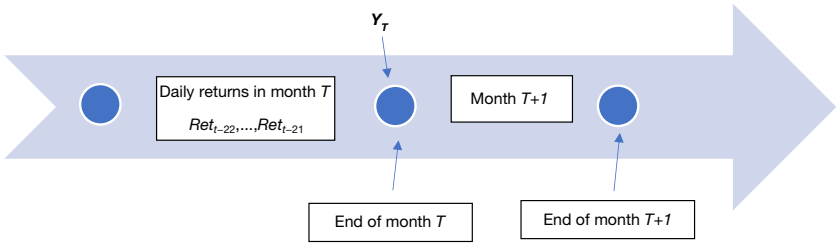
FIGURE 1

Timing of the Returns and Volatility Expectations in the MIDAS Regressions

Figure 1 provides a depiction of the timeline for the variables in the MIDAS regression,

$$Y_T = \mu_1 + \phi_1 \sum_{t=1}^{22} b_1(t;\theta) Ret_{T-t} + \gamma_1 Y_{T-1} + X_{T-1} + \mu_{i,T},$$

where Y_T is one of the volatility expectation measures ($ExpVol_T$), variance risk premium (VRP_T), change in volatility expectation measures ($\Delta ExpVol_T$), and measures of unexpected innovation in $ExpVol_T$ ($\varepsilon_{1,T}$); X_{T-1}, refers to i) Vol_{T-1}, lagged market volatility in the prior month $T-1$, ii) $Vol_{T-6,T-2}$, the average lagged monthly market volatility computed over months $T-6$ to $T-2$, and iii) VoV_{T-1}, the lagged volatility-of-volatility, defined as the standard deviation of the daily VIX index in month $T-1$. The coefficient $b_1(t;\theta)$ captures the weights on the lagged daily excess returns in predicting the dependent variables Y_T. The capital letter T refers to the monthly frequency variables, and the lowercase t refers to the daily frequency variable. In the timeline below, Y_T is formed at the beginning of month $T+1$ / end of month T, and reflects the expected future volatility in month $T+1$. So, for volatility in Apr. 2024, the expectation is formed at the end of Mar. 2024 / beginning of Apr. 2024. Ret_{T-t} is the daily return in month T (i.e., in Mar. 2024).



Ghysels et al. (2006), we use a Beta polynomial that has two parameters, $\theta = [\theta_1; \theta_2]$, for the weighting function:

$$b_1(t;\theta) = \frac{f\left(\frac{t}{t^{max}}, \theta_1, \theta_2\right)}{\sum_{j=1}^{22} f\left(\frac{j}{t^{max}}, \theta_1, \theta_2\right)}$$

The high-frequency variable includes 22 lagged daily excess returns, $t = [1, \ldots, 22]$, and $t^{max} = 22$. The function $f(.)$ is defined as $f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b)$, where $\beta(a,b)$ is the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. The weights of $b_1(t;\theta)$ and $b_2(t;\theta)$ are normalized, setting the sum of the weights over all lagged returns equal to 1.

If investors extrapolate past excess daily returns in forming expectations about future volatility,[13] then we expect the following results from the MIDAS regressions:

i) The return coefficient ϕ_1 should be significantly negative with $ExpVol_T$ as the dependent variable. This negative coefficient is consistent with the predictions of the Atmaz (2022) model and indicates that investors have a higher (lower) expectation of $ExpVol_T$ after observing lower (higher) past returns.

[13]This is motivated by the burgeoning literature in behavioral asset pricing focuses on investors' extrapolative beliefs in forming expectations. Greenwood and Shleifer (2014) and Barberis et al. (2015) show that investors condition on past returns when forming expectations about future returns. Extrapolative beliefs about future return expectation results in investors expecting lower (higher) future returns following lower (higher) past returns, with a greater weight on the recent past returns than the distant past returns. Da et al. (2021) study extrapolative beliefs about future returns in the cross-section and document consistent results.

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ii) The return coefficient ϕ_1 should be significantly negative with VRP_T as the dependent variable. That is, investors are willing to pay a higher price to insure against the higher perceived future volatility as a result of extrapolating past lower returns. Hence, VRP_T should be higher following past lower returns.

iii) The return coefficient ϕ_1 should be significantly negative with $\Delta ExpVol_T$ and $\varepsilon_{1,T}$ as the dependent variables. These negative relations suggest an overaction bias in the volatility expectations induced by the return extrapolation. Hence, following lower past returns, the change in volatility expectations ($\Delta ExpVol_T$) and unexpected shocks to volatility expectations become larger.

iv) Across all four MIDAS regressions, the contributions from the recent past returns to the dependent variables Y_T should be larger than those of the distant past returns, a distinct feature of extrapolative belief. Therefore, the weighting coefficient $b(t;\theta)$ should decay over the past 22 lags.

III. Data

We collect and merge data from several sources. Stock return data is from CRSP. For the aggregate market, we use the daily CRSP value-weighted returns minus the risk-free rate (30-day treasury bill rate). For the stock level analysis, we focus on the U.S. stocks (with stock codes 10 and 11) that have valid options data from OptionMetrics. The VIX index is obtained from the Chicago Board of Exchange (CBOE) over the sample period from Jan. 1996 to Dec. 2020 and captures the risk-neutral expected volatility of the highly liquid S&P 500 index options.[14] The survey data is collected from the websites of Graham and Harvey's CFO survey (https://www.richmondfed.org/cfosurvey/) and the Shiller Stock Market Crash Confidence survey (https://som.yale.edu/centers/international-center-for-finance/data/stock-market-confidence-indices/united-states).

We obtain 5-minute intraday prices for the S&P 500 index (SPX) from TickHistory. Following Andersen, Bollerslev, Diebold, and Evens (2001) and Bollerslev et al. (2009), 5-minute intraday logged returns are calculated during the normal trading hours from 9:30AM to 4:00PM. Thus, each day consists of 78 intraday 5-minute returns. The monthly realized volatility, $RVol_T$, is the summation of squared 5-minute returns and squared daily overnight (close-to-open) returns during the month T. The $RVol_T$ is used to calibrate the forecasting model for VOL_T^{DY} and VOL_T^{Corsi}. It is also the key variable in the autoregressive regression Equation (1) for estimating unexpected shocks to volatility expectations ε_T.

The implied volatility for the S&P 500 index options is obtained from the OptionMetrics volatility surface database. The volatility surface files contain daily interpolated implied volatilities for a range of standardized options with constant maturities and moneyness.[15] These standardized options are only included if there exists sufficient option price data to interpolate the values

[14]For more details about the VIX index and its computation, see vixwhite.pdf (cboe.com).

[15]The standardized options have constant maturity of 30, 60, 91, 122, 152, 182, 273, 365, 574, and 730 days, and absolute delta values from 0.2 to 0.8 with fixed increment of 0.05.

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accurately. We focus on 30-day constant maturity at-the-money options that have a delta of 0.5 for the call and −0.5 for the put options. This mitigates the concern that implied volatilities from out-of-money options may capture additional information beyond expected volatilities, such as jump risk, tail risk premium, and so forth At-the-money options are also the most actively traded and the most liquid (Muravyev and Pearson (2020)). The sample period covers 25 years, from Jan. 1996 to Dec. 2020.

For the $ExpVol_T$ measures, VIX_T and IV_T have 300 valid monthly observations from Jan. 1996. For $ExpVol_T$ under the physical measures, including VOL_T^{GARCH}, VOL_T^{DY}, and VOL_T^{Corsi}, the first observation starts in Jan. 2001 since the out-of-sample forecast requires a 60-month rolling window to calibrate the forecasting model. Figure 2 displays the time-series plot of these variables.

Graph A of Figure 2 plots the evolution of $ExpVol_T$ measures. First, all $ExpVol_T$ measures are highly correlated. Second, all $ExpVol_T$ measures tend to move closely with the $RVol_T$, which confirms the information content of the $ExpVol_T$. Third, VIX_T and IV_T are systematically higher than $RVol_T$ and other conditional physical measures. This gap reflects the variance risk premium (i.e., the risk-neutral expectation is higher than the physical expectation). Graphs B, C, and D plot the VRP_T, $\Delta ExpVol_T$, and ε_T, respectively. The consistent observation is that the variance risk premium (Graph B), the change in volatility expectation (Graph C), and the unexpected shocks to volatility expectations (Graph D) are all correlated across the different measures. It is also evident that both $\Delta ExpVol_T$ and $\varepsilon_{1,T}$ are mean-reverting and less persistent than $ExpVol_T$.

Table 1 presents the descriptive statistics for the key variables. Panel A shows that the mean (median) of annualized *VIX* is 0.204 (0.187), with the minimum (maximum) value of 0.095 (0.685). This number is larger than the mean (median) value of the monthly realized volatility, RVOL, of 0.126 (0.110), and other volatility expectation proxies. Again, the higher value of VIX reflects the variance risk premium. Panel B reports the various VRP measures. The mean and median values are positive across all measures. This is consistent with the literature that the investors are willing to pay a premium to insure against future uncertainty. Panel C reports results for $\Delta ExpVol_T$. While the mean and median are close to 0, it still has a significant negative AR(1) coefficient. In contrast, for the unexpected shocks, ε_T, reported in Panel D, the mean values across all proxies are 0, and AR(1) coefficients are statistically insignificant. These results suggest that ε_T captures random shocks to $\Delta ExpVol_T$. Examining the impact of return extrapolation on ε_T will inform us about the bias in $ExpVol_T$.

IV. Market-Level Return Extrapolation

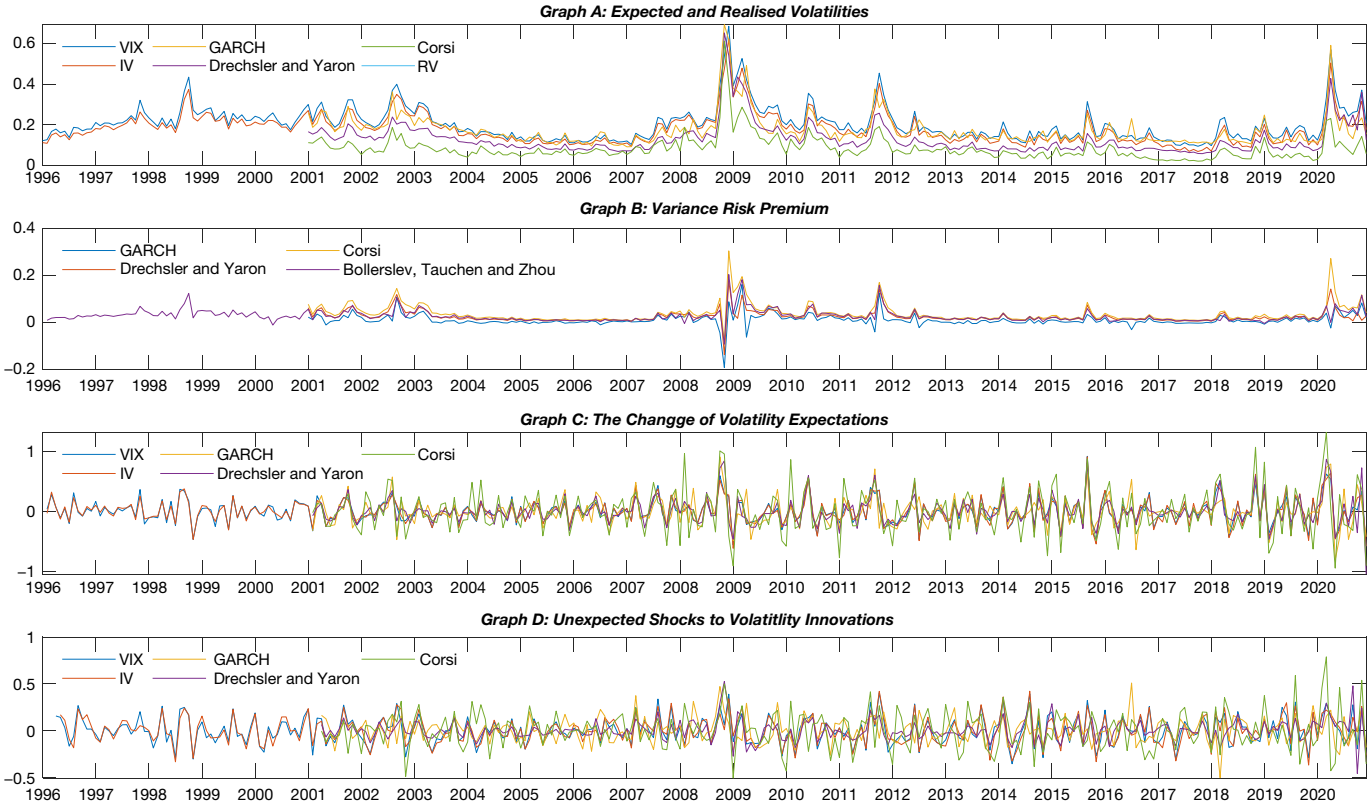
A. Return Extrapolation and Volatility Expectations

To visualize the relationship between the return in the month $T-1$ and $ExpVol_T$, we first plot monthly VIX_T against the prior month's aggregate market excess returns. Graph A of Figure 3 shows that when prior month returns are lower,

FIGURE 2

Time-Series Plots of Expected Volatility, Variance Risk Premium, the Change in Volatility Expectations, and Unexpected Shocks of Volatility Expectations

Figure 2 depicts time-series plots for volatility expectation measures ($ExpVol_T$) in Graph A, variance risk premium (VRP_T) in Graph B, the change in volatility expectation measures ($\Delta ExpVol_T$) in Graph C, and measures of unexpected innovation in $ExpVol_T$ ($\varepsilon_{1,T}$) in Graph D.



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TABLE 1

Descriptive Statistics

Table 1 Panel A reports descriptive statistics for the expected volatility. VIX_T is the VIX index observed at the beginning of month T. IV_T is the average implied volatility of 30-day at-the-money call and put S&P 500 index options observed at the beginning of the month T. VOL_T^{GARCH} is the 1-month-ahead forecasted conditional volatility from the GARCH(1,1) model. VOL_T^{DY} is the 1-month-ahead forecasted conditional volatility following Drechsler and Yaron (2011). VOL_T^{Corsi} is the 1-month-ahead forecasted conditional volatility from the heterogeneous autoregressive model of Corsi (2009). $RVOL_T$ is the monthly realized volatility observed at the end of the month. It is constructed using 5-minute squared returns during the normal trading hours from 9:30AM to 4:00PM and squared overnight (close-to-open) returns during the month T. All conditional volatilities, VOL_T^{GARCH}, VOL_T^{DY} and VOL_T^{Corsi} are estimated recursively with daily data from the past 60-month rolling window. Panel B reports descriptive statistics for the monthly variance risk premium (VRP_T). VRP_T is defined as the difference between variance under risk-neutral expectation and physical expectation. $VRP_T = \mathbb{E}_T^Q(Var_{T,T+1}) - \mathbb{E}_T^P(Var_{T,T+1})$, where $\mathbb{E}_T^Q(Var_{T,T+1}) = VIX_T^2$, and $\mathbb{E}_T^P(Var_{T,T+1})$ are based on different conditional volatility forecasting models. For the physical variance expectation $\mathbb{E}_T^P(Var_{T,T+1})$, VRP_T^{GARCH} uses 1-month-ahead forecasted conditional variance from the GARCH(1,1) model for; VRP_T^{DY} uses 1-month-ahead forecasted conditional variance following Drechsler and Yaron (2011); VRP_T^{Corsi} uses the 1-month-ahead forecasted conditional variance from the heterogeneous autoregressive model of Corsi (2009); and VRP_T^{BTZ} uses the month $T-1$ realized variance following Bollerslev, Tauchen, and Zhou (2009). Panel C reports descriptive statistics for the expected volatility innovations. The expected volatility innovation ($\Delta ExpVol_T$) is the monthly change in the expected volatility. Panel D reports descriptive statistics for the unexpected shocks to volatility expectations, which are the residuals ε_1 from the linear regression $\Delta ExpVol_T = \beta_0 + \beta_{1,i} \times \Delta ExpVol_{T-i} + \beta_{2,i} \times \Delta RVol_{T-j} + \varepsilon_{1,T}$ that regress $\Delta ExpVol_T$ on the lagged $\Delta ExpVol_{T-i}$ and lagged $\Delta RVOL_{T-j}$, where $\Delta ExpVol_T$ is the monthly change of the expected volatility as defined in Panel A and $\Delta RVOL_T$ is the monthly change of the realized volatility. ε_T^{VIX} is the unexpected shock to VIX_T. ε_T^{IV} is the unexpected shock to IV_T. ε_T^{GARCH} is the unexpected shock to VOL_T^{GARCH}. ε_T^{DY} is the unexpected shock to VOL_T^{DY} and ε_T^{Corsi} is the unexpected shock to VOL_T^{Corsi}. Reported descriptive statistics include the mean, median, minimum, maximum, standard deviation, skewness, and kurtosis. Also reported are the AR(1) coefficients and the corresponding t-statistics. The sample period is Jan. 2001 to Dec. 2020.

	Mean	Median	Min	Max	Std. Dev.	Skewness	Kurtosis	AR(1)	t-Stats
Panel A. Expected Volatility									
VIX_T	0.204	0.187	0.095	0.685	0.083	1.893	8.781	0.798	(22.92)
IV_T	0.180	0.164	0.068	0.623	0.076	1.773	8.563	0.804	(23.41)
VOL_T^{GARCH}	0.174	0.154	0.092	0.693	0.080	3.230	17.235	0.714	(15.79)
VOL_T^{DY}	0.130	0.106	0.058	0.653	0.076	3.181	17.131	0.804	(20.83)
VOL_T^{Corsi}	0.084	0.069	0.021	0.612	0.059	4.223	31.892	0.704	(15.25)
$RVOL_T$	0.126	0.110	0.038	0.617	0.073	2.895	16.394	0.700	(16.87)
Panel B. Variance Risk Premium									
VRP_T^{GARCH}	0.011	0.006	−0.193	0.162	0.026	−0.325	24.115	0.079	(1.22)
VRP_T^{DY}	0.025	0.017	−0.138	0.176	0.026	1.280	17.432	0.198	(3.12)
VRP_T^{Corsi}	0.037	0.025	−0.087	0.305	0.039	3.188	19.196	0.464	(8.09)
VRP_T^{BTZ}	0.027	0.021	−0.093	0.204	0.026	2.510	16.740	0.274	(4.90)
Panel C. The Change of Volatility Expectations									
ΔVIX_T	0.002	−0.003	−0.581	0.916	0.208	0.476	4.364	−0.141	(−2.43)
ΔIV_T	0.002	−0.015	−0.615	0.907	0.215	0.359	4.257	−0.160	(−2.75)
ΔVOL_T^{GARCH}	−0.002	−0.028	−0.811	0.907	0.244	0.524	4.529	−0.174	(−2.70)
ΔVOL_T^{DY}	−0.001	−0.021	−1.045	0.868	0.230	0.687	6.697	−0.113	(−1.67)
ΔVOL_T^{Corsi}	−0.003	−0.011	−0.945	1.314	0.364	0.404	3.749	−0.191	(−2.94)
Panel D. Unexpected Shocks of Volatility Expectations									
ε_T^{VIX}	0.000	−0.005	−0.357	0.406	0.147	0.240	2.810	−0.088	(−1.51)
ε_T^{IV}	0.000	−0.006	−0.370	0.421	0.145	0.153	2.930	−0.029	(−0.50)
ε_T^{GARCH}	0.000	−0.010	−0.506	0.509	0.134	0.357	4.199	−0.031	(−0.48)
ε_T^{DY}	0.000	−0.012	−0.461	0.526	0.113	0.570	6.943	−0.032	(−0.48)
ε_T^{Corsi}	0.000	−0.001	−0.509	0.789	0.200	0.343	3.740	−0.120	(−1.84)

VIX_T tends to become higher. With a quadratic polynomial model, the fitted curve (adjusted R^2 = 0.402) is downward sloping, especially when the prior month's returns are negative. Graph B shows a milder downward-sloping relation between the prior month's returns and VRP_T^{BTZ} (with an adjusted R^2 of 0.371). Graphs C and D of Figure 3 show a similar negative relation between the prior month's returns and

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ΔVIX_T and ε_T^{VIX}, suggesting that VIX_T may be biased upward when returns in the prior month are low. The adjusted R^2s of the quadratic polynomial fitted model for ΔVIX_T and ε_T^{VIX} are 0.045 and 0.071, respectively. All these plots indicate that investors' expectations of future volatilities are affected by past returns.

1. Impact of Return Extrapolation on Expected Volatility

We now formally examine whether return extrapolation affects the expectations of future volatility for the aggregate market. Our analysis goes beyond the level of volatility expectations and aims to understand how return extrapolation affects the variance risk premium and induces biases in the volatility expectations. We use the MIDAS model that regresses monthly $ExpVol_T$ of the S&P 500 index on the lagged daily excess returns. All proxies for $ExpVol_T$ are observed at the beginning of each month T.

Panel A of Supplementary Material Table IA1 reports the estimation results for a simple AR(1) model with the control variables. The AR(1) coefficient is positive and highly significant across all $ExpVol_T$ proxies. For example, the coefficient is 0.777 (0.767) for VIX_T $\left(VOL_T^{Corsi}\right)$, and the t-statistic is 13.49 (13.25). This result confirms the persistence in volatility that is well documented in the literature. The simple AR(1) model seems to capture the $ExpVol_T$ reasonably well, with all the adjusted R^2s above 0.56.

The MIDAS regression results are reported in Table 2 Panel A. In the MIDAS model, the scaling parameter ϕ_1 captures the overall relationship between the high-frequency lagged daily excess returns and the low-frequency monthly $ExpVol_T$. The two parameters $\theta = [\theta_1, \theta_2]$ in the Beta polynomial weighting function jointly determine the contributions of the lagged daily excess Ret_{T-t} to the monthly $ExpVol_T$.

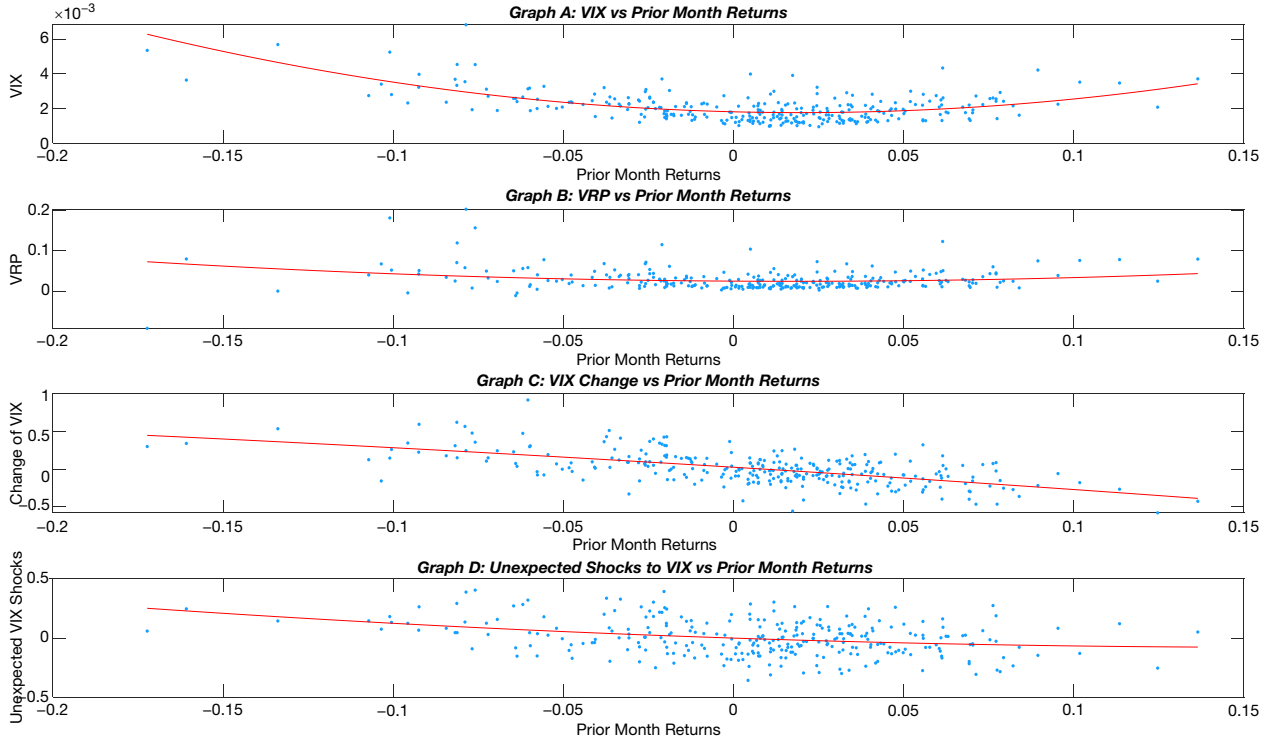
The coefficient ϕ_1 is negative and highly significant across all $ExpVol_T$ proxies. For the risk-neutral expected volatility (VIX_T), the coefficient is -0.565 (t-stat = -12.15). Similar results are found for the conditional physical measures. For example, the coefficient for VOL_T^{GARCH} is -0.442 (t-stat = -8.89). The negative coefficients point to an inverse relation between the monthly volatility expectations (under both risk-neutral and physical measures) and the past daily returns. In other words, lower (higher) past daily returns are associated with a higher (lower) expectation of future volatility. Moreover, the adjusted R^2 improves significantly, with the lowest value of 0.805 for VOL_T^{Corsi}.

Furthermore, the return extrapolation effect further stipulates that the recent past returns should have a greater impact than that of the distant past returns. The MIDAS estimation results confirm this. Graph A of Figure 4 shows the relative weights across 22 lagged excess daily returns for VIX_T. It is clear that contributions from past daily returns decay as we look into the more distant past. Specifically, the total weights from the most recent 10 days (15 days) are 59% (81%). A similar pattern obtains for other $ExpVol_T$ proxies. For example, the total weights from the most recent 10 days (15 days) are 57% (79%) and 46% (68%) for VOL_T^{IV} and VOL_T^{DY}, respectively. Thus, the negative relation between past returns and volatility expectation is dominated by the recent past returns. Further corroborative evidence from survey-based expectations is presented in Table 3.

FIGURE 3

Scatter Plots of VIX, Variance Risk Premium, Change of VIX, and Unexpected Shocks to VIX against Prior Month Returns

Figure 3 Graph A plots the VIX_T against the market excess returns in the prior month $T-1$. Graph B plots the change in VIX ΔVIX_T against the market excess returns in the prior month $T-1$. Graph C plots the VRP_T^{BTZ} (defined as $VIX_T^2 - RV_{T-1}^2$) against the market excess returns in the prior month $T-1$. Graph D plots the Unexpected Shocks to Volatility Expectations ε_T^{VIX} against the market excess returns in the prior month $T-1$.



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TABLE 2

MIDAS Regression Estimation Results

Table 2 reports model estimates for the MIDAS regressions that regress monthly dependent variable Y_T on 22 lagged daily excess returns, lagged dependent variable Y_{T-1}, and control variables. $Y_T = \mu_1 + \phi_1 \sum_{t=1}^{22} b_1(t;\theta) Ret_{T-t} + \gamma_1 Y_{T-1} + X_{T-1} + \tau_T$. In Panel A, the dependent variable is monthly expected volatility. In Panel B, the dependent variable is the variance risk premium. In Panel C, the dependent variable is the change in volatility expectations. In Panel D, the dependent variable is unexpected shocks to volatility expectations. The MIDAS regressions utilize a Beta polynomial with two parameters, $\theta = [\theta_1; \theta_2]$ for the weighting function. Each lagged return has a weight $(t;\theta) = \frac{f(\frac{t}{max}, \theta_1, \theta_2)}{\sum_{t=1}^{22} f(\frac{t}{max}, \theta_1, \theta_2)}$, with $t = [1, \ldots, 22]$, and $t^{max} = 22.f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b)$, and $\beta(a,b)$ is based on the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. Also reported is adjusted R^2 for each model. *T*-statistics are reported in parentheses. For the control variables, Vol_{T-1} is the lagged market volatility in the prior month $T-1$, $Vol_{T-6,T-2}$ is the average lagged monthly market volatility from $T-6$ to $T-2$, VoV_{T-1} is the lagged volatility-of-volatility defined as the standard deviation of the daily VIX index during the month $T-1$. Panel B reports model estimates for the MIDAS regressions that regress monthly expected volatility. Also reported is adjusted R^2 for each model. *t*-statistics are reported in parentheses.

Panel A. Expected Volatility

	Dependent Variable				
	VIX_T	IV_T	VOL_T^{GARCH}	VOL_T^{DY}	VOL_T^{Corsi}
Intercept	−0.631	−0.572	−0.993	−0.805	−0.943
	(−8.65)	(−7.78)	(−11.59)	(−9.57)	(−7.92)
Lagged Ret Coefficient ϕ_1	**−0.565**	**−0.632**	**−0.442**	**−0.628**	**−1.035**
	(−12.15)	**(−12.89)**	**(−8.89)**	**(−13.02)**	**(−11.98)**
Beta Polynomial θ_1	0.986	0.986	1.062	0.991	1.011
	(163.77)	(171.46)	(20.08)	(150.91)	(73.79)
Beta Polynomial θ_2	1.328	1.251	0.988	0.997	1.000
	(8.9)	(9.82)	(149.99)	(164.62)	(142.36)
AR(1)	0.682	0.726	0.520	0.679	0.679
	(18.76)	(21.23)	(12.59)	(20.12)	(17.26)
Vol_{T-1}	4.090	5.235	7.687	3.773	−4.768
	(1.7)	(2.1)	(3.26)	(1.81)	(−1.25)
$Vol_{T-6,T-2}$	7.954	6.518	6.759	11.187	10.945
	(2.94)	(2.35)	(2.48)	(3.95)	(2.66)
VoV_{T-1}	0.036	0.025	0.045	0.043	0.056
	(4.13)	(2.76)	(5.03)	(5.31)	(3.8)
Adj R^2	0.867	0.869	0.827	0.908	0.805

Panel B. Variance Risk Premium

	Dependent Variable			
	VRP_T^{GARCH}	VRP_T^{DY}	VRP_T^{Corsi}	VRP_T^{BTZ}
Intercept	0.002	0.012	0.011	0.012
	(0.95)	(4.56)	(3.83)	(5.46)
Lagged Ret Coefficient ϕ_1	**−0.043**	**−0.032**	**−0.040**	**−0.042**
	(−6.82)	**(−4.57)**	**(−5.03)**	**(−7.27)**
Beta Polynomial θ_1	0.964	0.981	0.973	0.983
	(133.77)	(90.46)	(107.82)	(148.63)
Beta Polynomial θ_2	1.877	2.110	2.032	2.658
	(5.31)	(4.54)	(5.34)	(6.93)
AR(1)	−0.080	0.032	0.022	−0.104
	(−1.61)	(0.51)	(0.36)	(−1.90)
Vol_{T-1}	−1.886	0.351	1.710	−0.458
	(−5.99)	(1.00)	(4.36)	(−1.50)
$Vol_{T-6,T-2}$	2.983	1.433	3.633	4.027
	(10.1)	(4.25)	(7.81)	(12.61)
VoV_{T-1}	0.004	0.004	0.003	0.004
	(2.69)	(2.77)	(2.56)	(1.43)
Adj R^2	0.485	0.310	0.608	0.547

Panel C. The Change in Volatility Expectations

	Dependent Variable				
	ΔVIX_T	ΔIV_T	ΔVOL_T^{GARCH}	ΔVOL_T^{DY}	ΔVOL_T^{Corsi}
Intercept	−0.029	−0.021	−0.028	−0.038	−0.016
	(−1.67)	(−1.16)	(−1.52)	(−2.34)	(−0.55)

(continued on next page)

TABLE 2 (continued)
MIDAS Regression Estimation Results

Panel C. The Change in Volatility Expectations (continued)

	Dependent Variable				
	ΔVIX_T	ΔIV_T	ΔVOL_T^{GARCH}	ΔVOL_T^{DY}	ΔVOL_T^{Corsi}
Lagged Ret Coefficient ϕ_1	**−0.590 (−11.86)**	**−0.639 (−12.41)**	**−0.495 (−8.98)**	**−0.646 (−12.91)**	**−1.096 (−12.19)**
Beta Polynomial θ_1	0.988 (147.39)	0.987 (146.75)	1.058 (17.98)	0.993 (138.21)	1.012 (72.06)
Beta Polynomial θ_2	1.213 (8.93)	1.160 (7.94)	0.991 (141.92)	0.997 (161.96)	0.999 (145.83)
AR(1)	−0.294 (−5.83)	−0.295 (−6.02)	−0.380 (−8.2)	−0.373 (−8.14)	−0.314 (−6.39)
Vol_{T-1}	1.336 (0.53)	2.238 (0.85)	4.497 (1.75)	2.923 (1.34)	−8.385 (−2.13)
$Vol_{T-6,T-2}$	−9.578 (−4.03)	−9.399 (−3.82)	−17.673 (−7.05)	−12.643 (−5.86)	−11.804 (−3.02)
VoV_{T-1}	0.038 (3.97)	0.032 (3.25)	0.046 (4.66)	0.048 (5.54)	0.061 (3.92)
Adj R^2	0.540	0.555	0.585	0.655	0.547

Panel D. Unexpected Shocks to Volatility Expectations

	Dependent Variable				
	ε_T^{VIX}	ε_T^{IV}	ε_T^{GARCH}	ε_T^{DY}	ε_T^{Corsi}
Intercept	−0.009 (−0.63)	−0.004 (−0.25)	0.013 (0.88)	0.003 (0.23)	0.023 (1.1)
Lagged Ret Coefficient ϕ_1	**−0.306 (−7.43)**	**−0.330 (−8.21)**	**−0.091 (−2.77)**	**−0.206 (−6.36)**	**−0.340 (−6.99)**
Beta Polynomial θ_1	0.987 (135.99)	0.986 (148.5)	1.173 (2.7)	0.992 (119.67)	1.677 (4.27)
Beta Polynomial θ_2	2.325 (6.56)	2.230 (7.13)	4.880 (1.92)	2.714 (4.88)	4.748 (3.6)
AR(1)	−0.111 (−1.84)	−0.043 (−0.72)	−0.092 (−1.45)	−0.098 (−1.57)	−0.113 (−1.8)
Vol_{T-1}	4.148 (1.93)	4.865 (2.35)	8.513 (4.13)	8.008 (4.77)	4.865 (1.62)
$Vol_{T-6,T-2}$	−0.837 (−0.43)	0.744 (0.39)	−5.349 (−2.85)	0.308 (0.21)	−0.416 (−0.15)
VoV_{T-1}	0.004 (0.45)	−0.003 (−0.36)	−0.010 (−1.24)	−0.012 (−1.89)	−0.013 (−1.17)
Adj R^2	0.268	0.288	0.126	0.259	0.210

While the above results suggest that return extrapolation impacts volatility expectations, two related questions are yet unanswered. First, does return extrapolation affect investors' willingness to pay to insure against future perceived (subjective) volatility? Second, does return extrapolation induce biases in $ExpVol_T$? We address these two questions next.

2. Impact of Return Extrapolation on the Variance Risk Premium

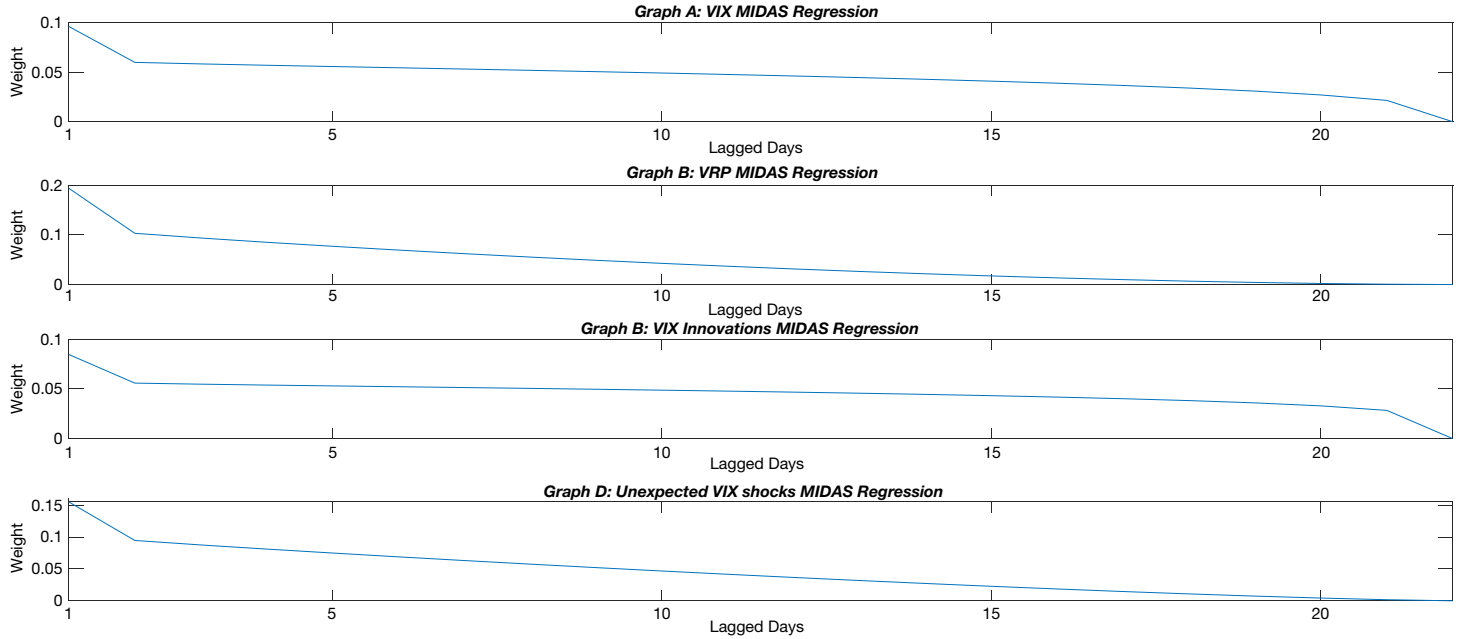
Carr and Wu (2009) argue that VRP is equivalent to the ex ante unit cost of hedging the variance risk. Hence, if investors extrapolate returns when forming volatility expectations, we expect that risk-averse investors would be willing to pay a higher price (higher VRP) to hedge against the higher perceived (subjective) risks after observing lower returns. To test this, we use the monthly VRP_T as the

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FIGURE 4

MIDAS Weights of Lagged Daily Returns

Figure 4 Graph A plots the relative weights contributed by the 22 lagged daily excess returns in the MIDAS regression with VIX_T as the dependent variable. Graph B plots relative weights contributed by the 22 lagged daily excess returns in the MIDAS regression with VRP_T^{BTZ} (defined as $VIX_T^2 - RV_{T-1}^2$) as the dependent variable. Graph C plots relative weights contributed by the 22 lagged daily excess returns in the MIDAS regression with the expected volatility innovations ΔVIX_T as the dependent variable. Graph D plots relative weights contributed by the 22 lagged daily excess returns in the MIDAS regression with the unexpected shocks to volatility expectations ε_T^{VIX} as the dependent variable. The weights are calculated by weighting function parameters in the MIDAS regression. The MIDAS regressions utilize a Beta polynomial with two parameters, $\theta = [\theta_1; \theta_2]$ for the weighting function. The weight of each lagged return is $(t;\theta) = \frac{f\left(\frac{t}{t^{max}}, \theta_1, \theta_2\right)}{\sum_{i=1}^{22} f\left(\frac{i}{t^{max}}, \theta_1, \theta_2\right)}$, with $t = [1, \ldots, 22]$, and $t^{max} = 22$. $f(z, a, b) = z^{a-1}(1-z)^{b-1}/\beta(a, b)$, and (a, b) is based on the Gamma function $\beta(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$.



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TABLE 3

Return Extrapolation Effects and Survey-based Volatility Expectation

Table 3 reports MIDAS regression estimates for survey-based expected volatility. CFO is the expected volatility taken from Graham and Harvey's CFO survey. Shiller Individual and Shiller Institution are the expected volatilities taken from the Shiller stock market survey. In Panel A, the dependent variable is the expected volatility level. In Panel B, the dependent variable is the change in the expected volatility. The MIDAS model estimates that regresses the survey-based volatility expectations on lagged daily excess returns over the prior quarter (6 months) and control variables for CFO (Shiller) survey volatility expectations. $Y_T = \mu_1 + \phi_1 \sum_{t=1}^{22} b_1(t;\theta) Ret_{T-t} + \gamma_1 Y + \tau_T$. The MIDAS regressions utilize a Beta polynomial with two parameters, $\theta = [\theta_1; \theta_2]$ for the weighting function. Each lagged return has a weight for $W(t;\theta) = \frac{f\left(\frac{t}{t^{max}}, \theta_1, \theta_2\right)}{\sum_{j=1}^{22} f\left(\frac{j}{t^{max}}, \theta_1, \theta_2\right)}$, *with* $t = [1, \ldots, t^{max}]$, $t^{max} =$ 66 (22*3) for the quarterly CFO survey, $t^{max} =$ 132 (22*6) for the 6-month moving average for Shiller's survey. $f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b)$, *and* $\beta(a,b)$ is based on the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. Also reported is adjusted R^2 for each model. *t*-statistics are reported in parentheses.

Panel A. Survey-Expected Volatility

	CFO	Shiller_Individual	Shiller_Institution
Intercept	0.025	0.042	0.078
	(5.08)	(2.85)	(4.38)
Lagged Ret Coefficient ϕ_1	**−0.013**	**−0.071**	**−0.048**
	(−2.62)	**(−4.34)**	**(−2.34)**
Beta Polynomial θ_1	2.075	1.033	2.186
	(2.16)	(13.91)	(1.40)
Beta Polynomial θ_2	3.531	1.862	6.692
	(1.65)	(3.51)	(1.03)
AR(1)	0.412	0.948	0.874
	(3.62)	(39.27)	(29.01)
Lagged Vol	0.383	−1.191	−0.617
	(2.64)	(−2.26)	(−0.77)
Lagged VoV	0.047	0.241	0.307
	(1.14)	(2.55)	(2.12)
R^2	0.477	0.914	0.837
Panel B. Survey Expected Volatility Change			
Intercept	0.001	0.011	0.009
	(0.38)	(2.87)	(1.79)
Lagged Ret Coefficient ϕ_1	**−0.020**	**−0.073**	**−0.051**
	(−2.42)	**(−4.31)**	**(−2.28)**
Beta Polynomial θ_1	2.025	1.032	1.535
	(1.79)	(13.88)	(2.15)
Beta Polynomial θ_2	6.350	1.820	2.618
	(1.11)	(3.71)	(1.33)
AR(1)	−0.320	0.050	−0.009
	(−2.99)	(0.78)	(−0.15)
Lagged Vol	−0.002	−1.373	−1.592
	(−0.01)	(−2.61)	(−2.17)
Lagged VoV	0.002	0.199	0.291
	(0.04)	(2.07)	(2.07)
R^2	0.236	0.119	0.045

dependent variable in the MIDAS regression and study its relation with past daily excess returns. As explained in Section II, we consider several alternatives VRP_T measures to ensure that the results are robust.

Panel B of Supplementary Material Table IA1 reports estimation results for a simple AR(1) model. Unlike $ExpVol_T$, the AR(1) coefficient is no longer statistically significant. The adjusted R^2 ranges between 0.131 $\left(VRP_T^{GARCH}\right)$ to 0.443 $\left(VRP_T^{Corsi}\right)$. The estimation results suggest that VRP_T is less persistent compared with the $ExpVol_T$. Nevertheless, we control for the lagged VRP_{T-1} in the MIDAS regressions.

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Panel B of Table 2 reports estimation results for the MIDAS regression. First, the lagged return coefficient ϕ_1 is negative and highly significant across all VRP_T measures. The coefficient is −0.043 (t-stat = −6.82), −0.032 (t-stat = −4.57), −0.040 (t-stat = −5.03), and −0.041 (t-stat = −7.27) for VRP_T^{GARCH}, VRP_T^{DY}, VRP_T^{Corsi}, and VRP_T^{BTZ}, respectively. The negative coefficients confirm that the lagged daily returns are inversely related to the subsequent monthly VRP. That is, investors are willing to pay a higher premium for hedging against the higher perceived variance risk following lower returns. Second, the adjusted R^2 improves substantially from the AR(1) model. For example, it increases from 0.131 (0.384) to 0.485 (0.547) for VRP_T^{GARCH} $\left(VRP_T^{BTZ}\right)$. Third, consistent with the extrapolative framework, the recent past returns have a larger impact than that of the distant past returns, supporting the return extrapolation interpretation.

Graph B of Figure 4 plots the contributions from the 22 lagged daily excess returns to VRP_T^{BTZ}. The total weight from the most recent 10 days (15 days) is 83% (96%). In stark contrast, the total weight is only a trivial 0.7% from the most distant 5 days and 7.6% from the most distant 10 days. Similarly, the total weights from the most recent 10 days (15 days) are 75% (92%), 76% (93%), and 76% (93%) for VRP_T^{GARCH}, VRP_T^{DY}, and VRP_T^{Corsi}, respectively.

In sum, the results in this section suggest that return extrapolation by investors impacts volatility expectations. The lower (higher) past returns lead to higher (lower) volatility expectations, which are accompanied by a higher (lower) contemporaneous premium that investors are willing to pay to hedge against the perceived or subjective expected volatility. We next proceed to test whether return extrapolation induces biases in volatility expectations in light of the predictions of Atmaz (2022).

3. Return Extrapolation and the Volatility Expectation Bias

Identifying return extrapolation-driven biases in $ExpVol_T$ is not straightforward due to the following reasons. First, Atmaz (2022) demonstrates that the subjective variance expectation, given the extrapolative beliefs $E_T^S[Var]$ is different from the rational variance expectation $E_T^R[Var]$, and the difference is due to the bias induced by return extrapolation. Unlike $E_T^R[Var]$, $E_T^S[Var]$ is affected by the degree of extrapolation and sentiment elasticity. However, $E_T^R[Var]$ is not observed. Second, the realized volatility is not an ideal proxy for the ex ante rational variance expectation $E_T^R[Var]$ due to well-known stylized facts about the volatility process, including leverage and persistence. Recently, Giglio and Kelly (2018) show that realized volatility is larger than what can be justified by the standard models. All of these issues create measurement hurdles to quantitatively measuring the volatility expectation bias.

Despite the measurement difficulty, the extrapolation bias can be examined indirectly using the change of volatility expectation ($\Delta ExpVol_T$) and unexpected shocks to the volatility expectations (ε_T). Atmaz (2022) shows that extrapolating lower (higher) past returns leads to higher (lower) volatility expectations with an overestimation (underestimation) bias.[16] If return extrapolation induces persistent

[16]Given that $E_T^R[Var]$ is not observed, we cannot directly measure the difference between $E_T^S[Var]$ and $E_T^R[Var]$. Hence, in this section, we conduct indirect tests that provide support for the impact of return

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and systematic biases to volatility expectations, we expect that extrapolating past lower (higher) returns results in an overestimation (underestimation) bias in the $ExpVol_T$. This overestimation (underestimation) should be associated with increases (decreases) in $ExpVol_T$. Hence, $\Delta ExpVol_T$ should exhibit a significant return extrapolation structure with a negative relationship with past returns. Similarly, the overestimation (underestimation) bias should be captured by the high (low) value of the unexpected shocks ε_T. A significant return extrapolation structure in $\Delta ExpVol_T$ and ε_T will provide evidence for the extrapolative bias in volatility expectations.

First, we calculate the change of $\Delta ExpVol_T$ and then use it as the dependent variable in the MIDAS regression. Supplementary Material Table IA1 Panel C presents the estimation results for a simple AR(1) model with control variables. The AR(1) coefficient is negative, suggesting a mean-reverting feature in $\Delta ExpVol_T$, which is distinct from the volatility persistence effect in $ExpVol_T$. The AR(1) coefficient, although negative, is insignificant for $\Delta VIX_T, \Delta VOL_T^{GARCH}$ *and* ΔVOL_T^{DY}. The adjusted R^2s range between 0.027 (ΔVIX_T) and 0.089 $\left(\Delta VOL_T^{GARCH}\right)$.

The MIDAS model estimation results are reported in Table 2 Panel C. Again, the lagged return coefficient ϕ_1 is negative and highly significant across all $\Delta ExpVol_T$ measures. The coefficients are in a range between −0.495 (t-stat = −8.98) for ΔVOL_T^{GARCH} and −1.096 (t-stat = −12.19) for ΔVOL_T^{Corsi}. The recent past returns have a larger impact than the distant past returns, which is the distinctive feature of return extrapolation. Graph C of Figure 4 shows the decay in the contribution across 22 lagged daily returns on ΔVIX_T. The total weight from the most recent 10 days (15 days) is 55% (78%). Similarly, the total weights from the most recent 10 days (15 days) are 54% (77%), 40% (64%), 46% (68%), and 44% (67%) for $\Delta IV_T, \Delta VOL_T^{GARCH}, \Delta VOL_T^{DY}$, and ΔVOL_T^{Corsi}, respectively. After incorporating the past daily returns, the adjusted R^2 improves significantly and ranges between 0.54 (ΔVIX_T) and 0.66 $\left(\Delta VOL_T^{DY}\right)$.

We estimate the unexpected shocks to $\Delta ExpVol_T$ based on the autoregressive model in Equation (1), which controls for the lagged changes in realized volatility, $\Delta RVOL_T$ and lagged change of expected volatility, $\Delta ExpVol_{T-i}$.[17] A higher (lower) value of ε_T reflects overreaction (underreaction) of $ExpVol_T$. We then use ε_T as the dependent variable in the MIDAS regression. Panel D of Supplementary Material Table IA1 reports the simple AR(1) model estimation results. The AR(1) coefficients and control variables are all statistically insignificant across all measures. The adjusted R^2 is either slightly negative or close to 0. These results confirm the randomness of the unexpected shocks.

The MIDAS estimation results are reported in Panel D of Table 2. The unexpected shocks exhibit a significant return extrapolation structure. The lagged return coefficients ϕ_1 are negative and highly significant across all ε_T measures. For example, the coefficients are −0.306 (t-stat = −7.43) for ε_T^{VIX} and −0.340

extrapolation on the bias in volatility expectations. In Section IV.A.4 we will use survey data to proxy for $E_T^S[Var]$ and directly test for the Atmaz (2022) result on the bias in volatility expectations.

[17]Using Bayesian Information Criterion (BIC), the optimal lag structure for the VAR model is three with the only exception being ΔVOL_T^{GARCH}, which has two lags.

(t-stat = −6.99) for ε_T^{Corsi}. Again, the contributions are larger for the recent lagged returns than those for the distant lagged returns. Graph D of Figure 4 shows the decay across the 22 lagged daily returns on ΔVIX_T. The total weight from the most recent 10 days (15 days) is 78% (94%). Similarly, the total weights from the most recent 10 days (15 days) are 76% (94%), 92% (99%), 82% (96%), and 86% (99%) for $\varepsilon_T, \varepsilon_T^{GARCH}, \varepsilon_T^{DY}$, and ε_T^{Corsi}, respectively. The adjusted R^2 improves substantially and ranges from 0.13 $\left(\varepsilon_T^{GARCH}\right)$ to 0.29 $\left(\varepsilon_T^{IV}\right)$.

4. Survey Evidence

Thus far, we have provided evidence about the return extrapolation effect on volatility expectations with both risk-neutral and physical expectation measures that are derived from market prices. In this section, we present corroborative evidence from survey-based volatility expectations. As outlined in Section II, there are limitations to these survey-based proxies. We present these results with the aforementioned caveats.

Panel A of Table 3 reports the MIDAS estimation results for survey-based volatility expectations. Since Graham and Harvey's CFO survey is conducted quarterly, and Shiller's Stock Market Crash Confidence Survey reports 6-month moving average results, we include lagged daily excess returns over the prior quarter and 6 months in the MIDAS regressions, respectively.[18] The lagged return coefficient ϕ_1 is negative and highly significant across both survey-based measures. Specifically, the coefficient is −0.013 (t-stat = −2.62) for the CFO survey and the Shiller stock market crash survey for individuals and institutions, and the coefficients are −0.071 (t-stat = −4.34), and −0.48 (−2.34), respectively. The coefficient estimates suggest that institutions may be less prone to extrapolation bias. Moreover, Shiller's stock market survey exhibits a decay from the recent to distant past returns.

We also find consistent results for the change of survey expected volatility. As reported in Panel B, the lagged return coefficient ϕ_1 is −0.020 (t-stat = −2.42) for the CFO survey. The coefficients are −0.073 (t-stat = −4.31) and −0.051 (−2.28) for the Shiller stock market crash survey for individuals and institutions, respectively. Collectively, the results of survey-based volatility expectations also suggest that return extrapolation leads to biased expectations of volatility.

B. Additional Findings and Potential Mechanisms

In this section, we first present additional empirical findings and then explore the potential mechanism that drives the extrapolative belief in volatility expectations.

1. Contemporaneous Relation between Return Expectations and Volatility Expectations.

Prior studies document that investors' return expectation is extrapolative and that lower (higher) past returns induce lower (higher) expectations for future returns

[18]Since the CFO survey is a quarterly measure, and Shiller's stock market survey is based on a 6-month rolling window result, we do not include Vol_{T-1} as the control variable and calculate VoV_{T-1} over the previous quarter and 6 months respectively

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TABLE 4

Contemporaneous Relation between Volatility Expectations and Return Expectations

Table 4 reports the regression results that regress volatility expectations on the contemporaneous survey-based return expectations. $ExpVol_T = \mu_1 + \beta_1 SurveyExpRet_T + Controls_T$. The volatility expectation includes the following proxies: VIX_T is the VIX index observed at the beginning of month T. IV_T is the average implied volatility of 30-day at-the-money call and put S&P 500 index options observed at the beginning of the month T. VOL_T^{GARCH} is the 1-month-ahead forecasted conditional volatility from the GARCH(1,1) model. VOL_T^{DY} is the 1-month-ahead forecasted conditional volatility following Drechsler and Yaron (2011). VOL_T^{Corsi} is the 1-month-ahead forecasted conditional volatility from the heterogeneous autoregressive model of Corsi (2009). CFO is the expected volatility taken from Graham and Harvey's CFO survey. Shiller_Individual and Shiller_Institution are the expected volatilities taken from Shiller's survey. The key explanatory variable is SurveyExpRet, which is the survey-based return expectation following (Cassella et al. (2022)). Also reported is adjusted R^2 for each model. t-statistics are reported in parentheses.

	VIX_T	IV_T	VOL_T^{GARCH}	VOL_T^{DY}	VOL_T^{Corsi}	CFO	Shiller_Individual	Shiller_Institution
Constant	0.276	0.306	0.361	0.566	0.235	0.202	0.199	0.046
	(1.70)	(1.91)	(1.69)	(2.71)	(0.87)	(5.66)	(6.03)	(6.23)
SurveyExpRet	**−5.246**	**−5.200**	**−8.784**	**−8.738**	**−10.635**	**−0.717**	**−0.762**	**−0.168**
	(−5.93)	**(−5.78)**	**(−9.04)**	**(−8.66)**	**(−6.70)**	**(−3.76)**	**(−5.17)**	**(−3.00)**
AR(1)	0.788	0.820	0.663	0.807	0.652	0.813	0.838	0.375
	(13.57)	(15.53)	(7.92)	(11.48)	(9.56)	(18.48)	(21.58)	(2.73)
Lagged Vol1	4.585	3.707	1.998	1.876	−1.730			
	(1.01)	(0.81)	(0.39)	(0.37)	(−0.22)			
Lagged Vol26	−3.388	−3.870	−1.402	−7.160	5.286	−1.169	−2.114	0.151
	(−0.72)	(−0.81)	(−0.3)	(−1.35)	(0.73)	(−1.06)	(−2.69)	(0.81)
Lagged VoV	−0.057	−0.056	−0.064	−0.063	−0.067	0.056	0.210	0.000
	(−3.78)	(−3.74)	(−3.94)	(−3.79)	(−2.56)	(0.25)	(1.27)	(0.00)
Adj R^2	0.600	0.638	0.593	0.710	0.557	0.716	0.805	0.324

(e.g., Greenwood and Shleifer (2014), Barberis et al. (2015), Da et al. (2021), and Cassella, Chen, Gulen, and Pekova (2022)). Combining these results with our main findings, we should expect a negative contemporaneous relation between return expectations and volatility expectations. With extrapolative belief when forming expectations, following lower (higher) returns, investors expect the future return to be lower (higher), while future volatility to be higher (lower). While this negative contemporaneous relation is at odds with the rational positive risk–return trade-off, it is supportive of the biased belief on return and volatility expectations induced by return extrapolation. This negative relation is obtained in Proposition 3 of Atmaz (2022).

Table 4 reports the results for the contemporaneous regression of volatility expectations ($ExpVol_T$) on return expectations. We use the monthly return expectations of Cassella et al. (2022) that are based on survey data and take into account the extrapolative belief in the return expectation formation process.[19] Across the columns, the dependent variables are different proxies for $ExpVol_T$. It is evident that there is a negative and significant contemporaneous relation between volatility expectations and return expectations. For example, the coefficient is −5.246 (t-stat = −5.93) for VIX_T, and −10.635 (t-stat = −6.70) for VOL_T^{Corsi}. This negative coefficient also exists with survey-based $ExpVol_T$. For example, it is −0.762 (t-stat = −5.17) and −0.168 (t-stat = −3.00) for the Shiller stock market crash survey for individuals and institutions and is marginally significant with a coefficient of −0.717 (t-stat = −3.76) for the CFO survey.

[19]We thank the authors for sharing the return expectation data.

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2. Return Extrapolation Degree and Volatility Expectation Bias

Cassella and Gulen (2018) show that a higher extrapolation degree (DOX) is associated with stronger return predictability of price-based mispricing variables, such as the D/P ratio. This result supports the model prediction of Barberis et al. (2015) that a higher DOX results in a quicker reversion of D/P, and hence a stronger subsequent mispricing correction effect. In the context of volatility expectation, Atmaz (2022) demonstrates that higher DOX results in greater sensitivity of stock price to fundamental variance shock, and hence induces more biased subjective volatility expectation. To test this novel model prediction of Atmaz (2022), we construct a conditional DOX measure following Cassella and Gulen (2018) and test the relationship between the conditional DOX_T and subsequent volatility expectation bias.

In each month T, we estimate the MIDAS regression with $ExpVol_T$ as the dependent variable, and calculate the total weights contributed by the most recent 5-days, dox_T.[20] A higher dox_T suggests greater impact from the recent than distant returns on $ExpVol_T$. Following Cassella and Gulen (2018), we estimate the five MIDAS regression with different estimation windows, including four rolling-windows that cover the previous 24 months, 36 months, 48 months, and 60 months, as well as an expanding window. The monthly DOX_T is the weighted average of dox_T with the inverse of the mean squared residuals (MSR) as the weights. Hence, a higher DOX_T reflects a higher conditional extrapolation degree. We then test the relationship between DOX_T and subsequent volatility expectation bias by estimating linear regression $Y_{T+1} = \mu_1 + \beta_1 DOX_T + Controls_T$. Since we are interested in the magnitude of expectation bias, we use the absolute value of the change of volatility expectation ($|\Delta ExpVol_{T+1}|$) and unexpected shocks to the volatility expectations ($|\varepsilon_{T+1}|$) as the dependent variable. The results are reported in Table 5.

In Panel A, the dependent variable is ($|\Delta ExpVol_{T+1}|$. The coefficient estimates of DOX_T are all positive and significant except for ΔVOL_T^{GARCH} and ΔVOL_T^{DY}. For example, the DOX_T coefficient is 1.575 (t-stat = 2.25) for $|\Delta VIX_T|$, and 12.224 (t-stat = 3.63) for $|\Delta VOL_T^{Corsi}|$. Table 5 Panel B shows consistent results with ($|\varepsilon_{T+1}|$) as the dependent variable. DOX_T coefficient is 0.278 (t-stat = 4.17) for $|\Delta VIX_T|$, and 3.898 (t-stat = 2.19) for $\Delta VOL_T^{Corsi}|$. The results confirm that when the extrapolation degree is higher, the magnitude of volatility expectation bias tends to be larger in the subsequent period. This evidence is indeed supportive of Propositions 2 and 3 of Atmaz (2022).

3. Extrapolating Past Returns Versus Past Realized Volatilities

We now explore potential mechanisms that drive investors' extrapolative beliefs. Due to constraints associated with survey volatility, the analysis focuses on the monthly risk-neutral and physical volatility expectations. We first investigate whether the extrapolative belief in volatility is indeed driven by extrapolating past returns but not by past realized volatilities. This exercise is crucial because the well-known persistence in volatility suggests that past volatility is a natural predictor of

[20] In MIDAS regression, each lagged return has a weight $(t;\theta) = \frac{f\left(\frac{t}{t^{max}},\theta_1,\theta_2\right)}{\sum_{j=1}^{22} f\left(\frac{j}{t^{max}},\theta_1,\theta_2\right)}$, with $t = [1,\ldots,22]$, and $t^{max} = 22, f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b)$, and $\beta(a,b)$ is based on the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$.

TABLE 5

Extrapolation Degree and Volatility Expectation Bias

Table 5 reports the results of the OLS regression $Y_T = \mu_1 + \beta_1 DOX_T + Controls_T$. DOX_T is a measure that captures the extrapolation degree when forming volatility expectations. Every month t, we estimate the MIDAS regression with $ExpVol_T$ as the dependent variable, and calculate the total weights contributed by the returns from the most recent 5-days dox_T. Following Cassella and Gulen (2018), we estimate the five MIDAS regression with different estimation windows, including four rolling-window periods that cover the previous 24 months, 36 months, 48 months, and 60 months, as well as an expanding window. For each MIDAS regression specification, the degree of extrapolation dox_T is the total weight contributed by the most recent 5-day lagged returns. The mean-squared-residuals (MSR), ($MSR_t^{24}, MSR_t^{36}, MSR_t^{48}, MSR_t^{60}, MSR_t^{expanding}$) is also calculated for each MIDAS specification. The monthly DOX_T is the weighted average of dox_T with the inverse of MSR as the weight. In Panel A, the dependent variable is the change in volatility expectations in absolute value, and in Panel B, the dependent variable is the unexpected shocks to volatility expectations in absolute value. t-statistics are reported in parentheses. The adjusted R^2 is also reported.

Panel A. Change in Volatility Expectations

	$\lvert\Delta VIX_T\rvert$	$\lvert\Delta IV_T\rvert$	$\lvert\Delta VOL_T^{GARCH}\rvert$	$\lvert\Delta VOL_T^{DY}\rvert$	$\lvert\Delta VOL_T^{Corsi}\rvert$
Intercept	0.295	0.027	0.702	0.862	2.017
	(1.71)	(0.13)	(3.58)	(2.78)	(2.04)
DOX_T	**1.575**	**4.348**	**−1.223**	**1.173**	**12.224**
	(2.25)	**(5.1)**	**(−0.66)**	**(0.62)**	**(3.63)**
AR(1)	0.018	0.035	0.157	−0.080	−0.039
	(0.26)	(0.48)	(2.17)	(−1.05)	(−0.51)
Vol_{T-1}	−45.619	−40.450	−39.176	−74.055	−77.973
	(−4.13)	(−2.83)	(−2.87)	(−3.85)	(−1.18)
$Vol_{T-6,T-2}$	−28.902	−38.170	4.771	−48.081	−37.694
	(−2.74)	(−2.79)	(0.37)	(−2.61)	(−0.59)
VoV_{T-1}	0.262	0.223	0.219	0.445	0.112
	(5.66)	(3.72)	(3.84)	(5.43)	(0.41)
Adj R^2	0.202	0.256	0.080	0.155	0.088

Panel B. Unexpected Shocks to Volatility Expectations

	$\lvert\varepsilon_T^{VIX}\rvert$	$\lvert\varepsilon_T^{IV}\rvert$	$\lvert\varepsilon_T^{GARCH}\rvert$	$\lvert\varepsilon_T^{DY}\rvert$	$\lvert\varepsilon_T^{Corsi}\rvert$
Intercept	0.333	0.404	0.694	0.291	0.953
	(2.56)	(2.48)	(5.57)	(1.85)	(1.78)
DOX_T	**2.278**	**2.548**	**1.336**	**2.294**	**3.898**
	(4.17)	**(3.95)**	**(1.12)**	**(2.26)**	**(2.19)**
AR(1)	−0.110	−0.048	−0.058	0.048	0.328
	(−1.46)	(−0.63)	(−0.77)	(0.61)	(4.56)
Vol_{T-1}	−13.467	−11.826	−10.421	−16.719	−24.663
	(−1.61)	(−1.05)	(−1.2)	(−1.65)	(−0.69)
$Vol_{T-6,T-2}$	−20.060	−23.505	−12.236	−11.265	−29.589
	(−2.48)	(−2.18)	(−1.46)	(−1.16)	(−0.86)
VoV_{T-1}	0.037	0.001	0.019	0.072	0.003
	(1.06)	(0.03)	(0.53)	(1.67)	(0.02)
Adj R^2	0.132	0.118	0.008	0.044	0.177

future volatility. To this end, we modify the MIDAS regression and include both lagged daily return and lagged daily realized volatility as the high-frequency variables. The results are reported in Table 6.

In Panel A, the dependent variable is $ExpVol_T$. For each $ExpVol_T$ proxy, we estimate two MIDAS models. Model 1 includes the lagged daily realized volatilities as the high-frequency variable. The coefficient on the lagged daily realized volatility, ϕ_2, is significantly positive. For example, the coefficient is 0.371 (t-stat = 13.14) for VIX_T and 0.374 (t-stat = 12.43) for VOL_T^{GARCH}. More importantly, in Model 2, when the MIDAS regression includes both the lagged daily return and the lagged daily realized volatility as the high-frequency variables, the lagged return coefficients ϕ_1 remain significantly negative across all $ExpVol_T$

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TABLE 6

Return Extrapolation Effects After Controlling for Daily Realized Volatilities

Table 6 reports results for MIDAS regression that include lagged daily realized volatility and lagged daily returns over the past 22 days as the high-frequency variables. In Panel A, the dependent variable is the monthly expected volatility. In Panel B, the dependent variable is the monthly variance risk premium. In Panel C, the dependent variable is the monthly change in volatility expectations. In Panel D, the dependent variable is the monthly unexpected shocks of volatility expectations. For each dependent variable, two MIDAS models are estimated. For Model 1, lagged daily realized volatility ($RealVol_{T-t}$) over the past month are included as the high-frequency variable. The MIDAS regression is $Y_T = \mu_1 + \phi_2 \sum_{t=1}^{22} b_2(t;\theta) RealVol_{T-t} + \gamma_1 Y_{T-1} + X_{i,T-1} + \varepsilon_{i,T}$. For model 2, both lagged daily realized volatility and lagged daily returns over the past 22 days as the high-frequency variables. The MIDAS regression is $Y_T = \mu_1 + \phi_1 \sum_{t=1}^{22} b_1(t;\theta) Ret_{T-t} + \phi_2 \sum_{t=1}^{22} b_2(t;\theta) RealVol_{T-t} + \gamma_1 Y_{T-1} + X_{i,T-1} + \varepsilon_{i,T}$. All MIDAS regressions control for the lagged market volatility in the prior month (Vol_{T-1}), the average lagged monthly market volatility during $T-6$ to $T-2$ ($Vol_{T-6,T-2}$), and the lagged volatility-of-volatility (VoV_{T-1}) defined as the standard deviation of the daily VIX index during the month $T-1$. The MIDAS regressions utilize a Beta polynomial with two parameters, $\theta = [\theta_1; \theta_2]$ for the weighting function. Each lagged return has a weight $W(t;\theta) = \frac{f(\frac{t}{t^{max}}, \theta_1, \theta_2)}{\sum_{j=1}^{23} f(\frac{j}{t^{max}}, \theta_1, \theta_2)}$, $with t = [1, \ldots, 22], t^{max} = 22. f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b), and \beta(a,b)$ is based on the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. Also reported is adjusted R^2 for each model. *t*-statistics are reported in parentheses.

Panel A. Expected Volatility

	Dependent Variable							
	VIX_T		VOL_T^{GARCH}		VOL_T^{DY}		VOL_T^{Corsi}	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Lagged Ret Coefficient ϕ_1		**−0.261**		**−0.231**		**−0.181**		**−0.136**
		(−4.07)		**(−4.94)**		**(−3.95)**		**(−3.15)**
Lagged Realized Vol	0.371	0.324	0.374	0.251	0.330	0.418	0.836	0.755
Coefficient ϕ_2	(13.14)	(7.25)	(12.43)	(7.01)	(8.83)	(11.25)	(24.78)	(20.17)
Adj R^2	0.867	0.917	0.857	0.831	0.902	0.938	0.925	0.930
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Panel B. Variance Risk Premium

	Dependent Variable							
	VRP_T^{GARCH}		VRP_T^{DY}		VRP_T^{Corsi}		VRP_T^{BTZ}	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Lagged Ret Coefficient ϕ_1		**−0.052**		**−0.028**		**−0.045**		**−0.048**
		(−8.51)		**(−3.77)**		**(−5.69)**		**(−9.10)**
Lagged Realized Vol Coefficient	0.013	0.006	0.022	0.022	0.029	0.023	0.017	0.011
ϕ_2	(2.82)	(1.54)	(5.00)	(5.12)	(5.25)	(4.71)	(4.63v)	(3.50)
Adj R^2	0.159	0.476	0.308	0.537	0.574	0.729	0.424	0.689
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C. The Change of Volatility Expectations

	Dependent Variable							
	ΔVIX_T		ΔVOL_T^{GARCH}		ΔVOL_T^{DY}		ΔVOL_T^{Corsi}	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Lagged Ret Coefficient ϕ_1		**−0.524**		**−0.106**		**−0.027**		**−0.881**
		(−10.72)		**(−3.04)**		**(−2.20)**		**(−10.59)**
Lagged Realized Vol	0.173	0.068	0.182	0.153	0.197	0.168	0.440	−0.214
Coefficient ϕ_2	(5.56)	(2.85)	(5.74)	(4.84)	(6.68)	(6.10)	(10.29)	(−4.98)
Adj R^2	0.335	0.617	0.474	0.532	0.477	0.463	0.464	0.533
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel D. Unexpected Shocks of Volatility Expectations

	Dependent Variable							
	ε_T^{VIX}		ε_T^{GARCH}		ε_T^{DY}		ε_T^{Corsi}	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Lagged Ret Coefficient ϕ_1		**−0.294**		−0.058		**−0.189**		**−0.183**
		(−7.50)		(−1.23)		**(−4.56)**		**(−3.36)**
Lagged Realized Vol Coefficient	0.059	0.008	0.025	0.031	−0.024	0.007	0.276	0.232
ϕ_2	(2.53)	(0.39)	(1.09)	(1.34)	(−1.38)	(0.38)	(10.53)	(8.00)
Adj R^2	0.063	0.432	0.004	−0.009	0.007	0.146	0.340	0.369
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

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proxies. The coefficients are −0.261 (*t*-stat = −4.07) for VIX_T, −0.231 (*t*-stat = −4.94) for VOL_T^{GARCH}, −0.181 (*t*-stat = −3.95) for VOL_T^{DY}, and −0.136 (*t*-stat = −3.15) for VOL_T^{Corsi}. Hence, the return extrapolation effect on volatility expectations is robust to controlling for the lagged daily realized volatility.

Similarly, the lagged return coefficients are significantly negative for VRP_T (Panel B), $\Delta ExpVol_T$ (Panel C), and $\varepsilon_{1,T}$ (Panel D) after controlling for lagged daily realized volatility.[21] For instance, the lagged return coefficients ϕ_1 in Model 2 are −0.048 (*t*-stat = −9.10) for VRP_T^{BTZ}, −0.524 (*t*-stat = −10.72) for ΔVIX_T, and −0.294 (*t*-stat = −7.50) for ε_T^{VIX} in Panels B, C, and D, respectively. These results paint a consistent picture that the impact of return extrapolation on volatility is distinct from extrapolating past realized volatility.

4. Investor Sentiment and Extrapolative Belief

Cassella and Gulen (2018) show that the extrapolation of returns is associated with investor sentiment, stock market volatility, and return salience. Han (2008) documents that the bearish sentiment due to low past returns leads to a more negative risk-neutral skewness of index options, which rational option pricing models cannot explain. Hence, we expect the impact of return extrapolation on volatility expectations to be more pronounced following investor pessimism. To test this investor sentiment mechanism, we first compare the effects of extrapolating past negative and positive returns on volatility expectations. If the impact of return extrapolation is driven more by pessimism, we expect the extrapolative belief in volatility to be asymmetric and more prominent due to past negative returns. Accordingly, we adjust the MIDAS regression and interact the high-frequency daily returns with the dummy variable D_{T-t}^{Neg} (D_{T-t}^{Pos}). The dummy variable D_{T-t}^{Neg} (D_{T-t}^{Pos}) equals 1 if the excess return on day $T - t$ is negative (positive), and 0 otherwise. This bi-variate MIDAS regression setup allows us to pin down the impact of past negative versus positive returns on volatility expectations and test whether it is the extrapolation of past negative returns that dominates. The results are reported in Table 7.

This table is organized in a format similar to that of Table 6. $ExpVol_T$ is the dependent variable in Panel A, while VRP_T is the dependent variable in Panel B. The results for $\Delta ExpVol_T$ and ε_T are reported in Panels C and D, respectively. The coefficients for the negative return interaction terms (Model 1) are significantly and consistently negative across all panels and robust across all proxies within each panel. For instance, the coefficients for ϕ^{Neg} are −0.592 (*t*-stat = −13.81) and −0.607 (*t*-stat = −13.30) for VIX_T and VOL_T^{DY}, respectively in Panel A, −0.023 (*t*-stat = −10.00) and −0.051 (*t*-stat = −13.80) for VRP_T^{Corsi} and VRP_T^{BTZ}, respectively in Panel B, −0.540 (*t*-stat = −11.50) and −0.688 (*t*-stat = −9.08) for ΔVIX_T and ΔVOL_T^{Corsi}, respectively in Panel C, and −0.264 (*t*-stat = −8.22) and −0.407 (*t*-stat = −7.97) for ε_T^{VIX} and ε_T^{Corsi}, respectively, in Panel D.

In contrast, the coefficients ϕ^{Pos} (Model 2) are much weaker, with a smaller magnitude and less statistical significance. For example, the coefficients are 0.130 (*t*-stat = 1.27) for VOL_T^{Corsi} in Panel A, −0.086 (*t*-stat = −0.94) for ΔVOL_T^{GARCH} in

[21] The only exception is ε_T^{GARCH} in Panel D, for which ϕ_1 *and* ϕ_2 are insignificant in Model 1 and Model 2.

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TABLE 7

Asymmetric Effects on Extrapolative Belief in Volatility Expectations

Table 7 reports results for MIDAS regression that include past daily positive excess returns and the past daily negative excess returns as separate high-frequency variables. In Panel A, the dependent variable is the monthly expected volatility. In Panel B, the dependent variable is the monthly variance risk premium. In Panel C, the dependent variable is the monthly change in volatility expectations. In Panel D, the dependent variable is the monthly unexpected shocks to volatility expectations. For each dependent variable, two MIDAS models are estimated. In Model 1, the MIDAS regression includes past negative excess returns as the high-frequency variable. $Y_T = \mu + \phi^{Neg}\sum_{t=1}^{22} b(t;\theta^{Neg}) * \left[Ret_{T-t} * D_{T-t}^{Neg}\right] + \phi^{Pos}\sum_{t=1}^{22} b(t;\theta^{Pos}) * \left[Ret_{T-t} * D_{T-t}^{Pos}\right] + \gamma * AR(1) + X_{j,T-1} + \varepsilon_T$, where $D_{T-t}^{Neg}(D_{T-t}^{Pos})$ is a dummy variable that equals 1 if Ret_{T-t} on day $T-t$ is negative (positive). All MIDAS regressions control for the lagged market volatility in the prior month (Vol_{T-1}), the average lagged monthly market volatility during $T-6$ to $T-2$ ($Vol_{T-6,T-2}$), and the lagged volatility-of-volatility defined as the standard deviation of the daily VIX index during the month $T-1$. (VoV_{T-1}. ϕ^{Neg} (ϕ^{Pos}) captures the relation between the dependent variable and lagged negative (positive) daily returns. The Beta polynomial is adopted for the weighting function. $\theta^{Neg} = \left[\theta_1^{Neg}, \theta_2^{Neg}\right]$ and $\theta^{Pos} = \left[\theta_1^{Pos}, \theta_2^{Pos}\right]$ determine the contribution of each lagged daily negative and positive excess returns respectively. The weight of each lagged return is $(t;\theta) = \frac{f\left(\frac{t}{t^{max}}, \theta_1, \theta_2\right)}{\sum_{j=1}^{22} f\left(\frac{j}{t^{max}}, \theta_1, \theta_2\right)}$, with $t = [1, \ldots, 22]$, $t^{max} = 22$. $f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b)$, and $\beta(a,b)$ is based on the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. Also reported is the adjusted R^2 for each model. *t*-statistics for the estimated coefficients are reported in parentheses.

Panel A. Expected Volatility

	Dependent Variable			
	VIX_T	VOL_T^{GARCH}	VOL_T^{DY}	VOL_T^{Corsi}
Negative Ret Coefficient ϕ^{Neg}	**−0.592**	**−0.450**	**−0.607**	**−0.899**
	(−13.81)	**(−11.47)**	**(−13.30)**	**(−10.31)**
Positive Ret Coefficient ϕ^{Pos}	−0.240	0.460	−0.148	0.130
	(−3.425)	(9.05)	(−1.67)	(1.27)
Adj R^2	0.893	0.897	0.906	0.809
Controls	Yes	Yes	Yes	Yes

Panel B. Variance Risk Premium

	Dependent Variable			
	VRP_T^{GARCH}	VRP_T^{DY}	VRP_T^{Corsi}	VRP_T^{BTZ}
Negative Ret Coefficient ϕ^{Neg}	**−0.019**	**−0.02**	**−0.023**	**−0.051**
	(−7.93)	**(−10.07)**	**(−10.00)**	**(−13.80)**
Positive Ret Coefficient ϕ^{Pos}	0.010	0.009	0.011	0.007
	(3.33)	(3.797)	(3.95)	(3.84)
Adj R^2	0.369	0.570	0.744	0.741
Controls	Yes	Yes	Yes	Yes

Panel C. The Change of Volatility Expectations

	Dependent Variable			
	ΔVIX_T	ΔVOL_T^{GARCH}	ΔVOL_T^{DY}	ΔVOL_T^{Corsi}
Negative Ret Coefficient ϕ^{Neg}	**−0.540**	**−0.511**	**−0.444**	**−0.688**
	(11.50)	**(−8.55)**	**(−9.52)**	**(−9.08)**
Positive Ret Coefficient ϕ^{Pos}	−0.555	−0.086	−0.255	−0.558
	(−7.76)	(−0.94)	(−**6**.11)	(−6.52)
Adj R^2	0.619	0.585	0.632	0.552
Controls	Yes	Yes	Yes	Yes

Panel D. Unexpected Shocks of Volatility Expectations

	Dependent Variable			
	ε_T^{VIX}	ε_T^{GARCH}	ε_T^{DY}	ε_T^{Corsi}
Negative Ret Coefficient ϕ^{Neg}	**−0.264**	**−0.195**	**−0.234**	**−0.407**
	(−8.22)	**(−4.98)**	**(−5.62)**	**(−7.97)**
Positive Ret Coefficient ϕ^{Pos}	−0.288	−0.007	−0.142	−0.188
	(−6.24)	(−0.43)	(−2.60)	(−3.56)
Adj R^2	0.452	0.145	0.188	0.274
Controls	Yes	Yes	Yes	Yes

Panel C, and −0.007 (*t*-stat = −0.43) for ε_T^{GARCH}. Overall, as compared to the past positive returns, past negative returns have a larger and more consistent impact on volatility expectations, indicating that extrapolative beliefs in volatility expectations are mainly driven by investors' pessimistic sentiment.

To further investigate the role of investor sentiment on volatility expectations, we split the sample into two subsamples based on whether the investor sentiment is pessimistic or optimistic. The idea is to test whether the impact of return extrapolation on volatility expectation is stronger following pessimistic periods. We classify month $T-1$ as pessimistic or optimistic depending on whether the excess monthly market return in the prior month is negative or positive. Recall that the dependent variable is as of month *T*. Hence, we are examining the impact of sentiment in month $T-1$ on the subsequent return extrapolation effect on volatility expectations.

When using negative and positive market returns in the prior month as the threshold, 104 and 183 months are classified as pessimistic and optimistic periods, respectively. Table 8 shows that the return coefficients ϕ_1 are significantly negative in the pessimistic subsample across all panels. Further, the magnitude of the return coefficients in the pessimistic subsample is almost always much larger than that of the optimistic subsample. For example, the coefficient for VIX_T is −0.669 (*t*-stat = −10.96) for pessimistic subsample versus −0.321 (*t*-stat = −4.58) for optimistic subsample in Panel A.[22]

Why is there an asymmetric impact on volatility expectations? Are investors more attentive, or do they react more to negative market returns? We find that the trading activity of out-of-the-money (OTM) put options is significantly higher than that of OTM call options in the month following negative returns. This suggests that investors are indeed affected more by losses than gains, as implied by the prospect theory of Kahneman and Tversky (1979).[23]

In addition, consistent with the model predictions of Atmaz (2022) the returns are more negatively skewed during the pessimistic periods. The mean (median) daily return skewness during month $T-1$ is −0.096 (−0.027) for the pessimistic sentiment subsample, while the mean (median) daily return skewness is −0.06 (0.07) for the optimistic sentiment subsample.

Atmaz (2022) shows that biased volatility expectations induced by return extrapolation impact derivative prices. This is consistent with investors' hedging demand being affected by extrapolative beliefs. Based on the evidence thus far, we expect investors' hedging demand to be negatively related to past returns and stronger following pessimistic periods. Following the prior literature, we use the monthly change in the OTM put option implied volatility (ΔIV_T^{OTMPut}) to proxy for investors' hedging demand and use it as the dependent variable in the MIDAS regressions.[24]

[22]In Supplementary Material Table IA2, we use the median excess returns over a 60-month rolling window as an alternative threshold to define pessimistic and optimistic periods and find similar results. With the alternative threshold, 139 (148) months are classified as pessimistic (optimistic).

[23]The OTM put-to-call trading volume ratios are 2.05 and 1.91 following pessimistic and optimistic months, respectively. The difference is statistically significant at the 1% level.

[24]See Gârleanu, Pedersen, and Poteshman (2009) and Bollen and Whaley (2004).

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TABLE 8

Extrapolative Belief in Pessimistic and Optimistic Periods

Table 8 reports the MIDAS regression results for pessimistic and optimistic subsamples. Month T is classified as pessimistic (optimistic) if the excess return in the prior month $T-1$ is negative (positive). MIDAS regressions are estimated for the pessimistic and optimistic subsamples separately. In Panel A, the dependent variable is the monthly expected volatility. In Panel B, the dependent variable is the monthly variance risk premium. In Panel C, the dependent variable is the monthly change in volatility expectations. In Panel D, the dependent variable is the monthly unexpected shocks to volatility expectations. All MIDAS regressions control for the lagged market volatility in the prior month (Vol_{T-1}), the average lagged monthly market volatility from $T-6$ to $T-2$ ($Vol_{T-6,T-2}$), and the lagged volatility-of-volatility (VoV_{T-1}) defined as the standard deviation of the daily VIX index during the month $T-1$. The MIDAS regressions utilize a Beta polynomial with two parameters, $\theta=[\theta_1;\theta_2]$ for the weighting function. Each lagged return has a weight $(t;\theta)=\frac{f\left(\frac{t}{22^{max}},\theta_1,\theta_2\right)}{\sum_{i=1}^{22} f\left(\frac{i}{22^{max}},\theta_1,\theta_2\right)}$, with $t=[1,\ldots,22]$, and $t^{max}=22.f(z,a,b)=z^{a-1}(1-z)^{b-1}/\beta(a,b)$, and $\beta(a,b)$ is based on the Gamma function $\beta(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$. Skewness_Mean (Skewness_Median) is the mean (median) value of daily return skewness in the prior month $T-1$ for each subsample. Also reported is the adjusted R^2 for each model. t-statistics for the estimated coefficients are reported in parentheses.

Panel A. Expected Volatility

	Dependent Variable							
	VIX_T		VOL_T^{GARCH}		VOL_T^{DY}		VOL_T^{Corsi}	
	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism
Lagged Ret Coefficient ϕ_1	−0.669	−0.321	−0.437	−0.197	−0.750	−0.440	−1.141	−0.488
	(−10.96)	(−4.58)	(−6.12)	(−3.57)	(−10.34)	(−7.33)	(−9.79)	(−3.94)
Adj R^2	0.899	0.863	0.856	0.849	0.912	0.929	0.814	0.796
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel B. Variance Risk Premium

	Dependent Variable							
	VRP_T^{GARCH}		VRP_T^{DY}		VRP_T^{Corsi}		VRP_T^{BTZ}	
	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism
Lagged Ret Coefficient ϕ_1	−0.060	−0.004	−0.041	−0.006	−0.050	−0.016	−0.055	−0.018
	(−5.38)	(−3.68)	(−3.03)	(−1.65)	(−3.17)	(−3.48)	(−5.19)	(−4.26)
Adj R^2	0.591	0.614	0.361	0.691	0.619	0.878	0.640	0.779
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C. The Change of Volatility Expectations

	Dependent Variable							
	ΔVIX_T		ΔVOL_T^{GARCH}		ΔVOL_T^{DY}		ΔVOL_T^{Corsi}	
	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism
Lagged Ret Coefficient ϕ_1	−0.710	−0.375	−0.522	−0.271	−0.738	−0.481	−1.224	−0.767
	(−11.01)	(−4.89)	(−6.07)	(−3.98)	(−10.31)	(−7.08)	(−9.91)	(−5.8)
Adj R^2	0.666	0.536	0.638	0.660	0.750	0.677	0.639	0.554
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel D. Unexpected Shocks of Volatility Expectations

	Dependent Variable							
	ε_T^{VIX}		ε_T^{GARCH}		ε_T^{DY}		ε_T^{Corsi}	
	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism	Pessimism	Optimism
Lagged Ret Coefficient ϕ_1	−0.312	−0.113	−0.021	−0.116	−0.195	−0.145	−0.386	−0.376
	(−6.6)	(−3.05)	(−1.88)	(−2.08)	(−5.43)	(−4.23)	(−5.31)	(−4.55)
Adj R^2	0.439	0.183	0.224	0.143	0.439	0.234	0.347	0.165
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The results in Table 9 confirm our conjecture. The past returns are negatively associated with ΔIV_T^{OTMPut} for the whole sample period (column 1), and this negative relation is much stronger following the pessimistic periods (column 2) as compared to the optimistic periods. Further, the return extrapolation effect on ΔIV_T^{OTMPut} is induced by extrapolating past negative returns (column 4) and past positive returns (column 5). The result holds after controlling for past realized volatility (column 6).

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TABLE 9

MIDAS Regression of Change in Implied Volatility for Out-of-Money Put Option

Table 9 reports the MIDAS regression results for out-of-money put option implied volatility change. In column, the MIDAS regression is estimated with whole sample data. In columns 2 and 3, the MIDAS regression is estimated with the pessimistic and optimistic subsample data. Month T is classified as pessimistic (optimistic) if the excess return in the prior month $T-1$ is negative (positive). MIDAS regressions are estimated for the pessimistic and optimistic subsamples separately. In columns 4 and 5, the MIDAS regressions include past daily positive excess returns and the past daily negative excess returns as separate high-frequency variables. In column 6, the MIDAS regression includes lagged daily realized volatility and lagged daily returns over the past 22 days as the high-frequency variables. All MIDAS regressions control for the lagged market volatility in the prior month (Vol_{T-1}), the average lagged monthly market volatility from $T-6$ to $T-2$ ($Vol_{T-6,T-2}$), and the lagged volatility-of-volatility (VoV_{T-1}) defined as the standard deviation of the daily VIX index during the month $T-1$. Also reported is the adjusted R^2 for each model. t-statistics for the estimated coefficients are reported in parentheses.

	Whole Sample	Pessimistic	Optimistic	Lagged Negative Returns	Lagged Positive Returns	Lagged Return and Lagged RVOL
	1	2	3	4	5	6
Lagged Ret Coef	−0.612	−0.706	−0.439			−0.540
	(−12.42)	(−11.16)	(−5.69)			(−11.31)
Negative Ret Coef				−0.569		
				(−12.17)		
Positive Ret Coef					−0.580	
					(−8.17)	
Lagged RVOL Coef						0.069
						(2.96)
Adj R	0.552	0.677	0.534	0.615	0.615	0.619

These results suggest that the extrapolative bias in volatility expectations is primarily induced by pessimistic investor sentiment. This finding complements the evidence about the mechanisms that drive extrapolative beliefs in returns. Barberis et al. (2018) argue that extrapolating past good news about the fundamentals could result in asset bubbles. In contrast, our results suggest that investor pessimism and the extrapolation of past negative returns contribute to biased beliefs about volatility expectations.

V. Return Extrapolation Effect on Volatility Expectations at the Stock Level

In this section, we shift our focus from the aggregate stock market to individual stocks. We find that the return extrapolation effect on volatility expectations exists at the stock level as well. Investors extrapolate past stock returns when forming expectations of volatility. This extrapolative belief induces an overestimation bias in volatility following low stock returns. As in the case of the aggregate market, stock-level volatility expectations are also predominantly driven by past negative stock returns. We also find that the return extrapolation effect on volatility expectations is more prevalent among firms with better visibility and more liquid stocks and options.

A. Results from the Stock-Level MIDAS Regressions

The key dependent variables are the stock level measures, $ExpVol_{iT}$, VRP_{iT}, $\Delta ExpVol_{iT}$, and $\varepsilon_{i,T}$. We follow Bakshi, Kapadia, and Madan (2003) to construct the model-free risk-neutral volatility (RNV) at the stock level. Specifically, we use the OptionMetrics volatility surface data and focus on the standardized

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options with a constant maturity of 30 days. At the beginning of each month T, the RNV is used as the proxy for the firm i's $ExpVol_{i,T}$ in month T. We follow Bollerslev et al. (2009) and calculate

$$VRP_{i,T} = RNV^2_{i,T} - RVOL^2_{i,T-1}.$$

$\Delta ExpVol_{i,T}$ is the monthly change of the expected volatility. Finally, the unexpected volatility innovations are the residuals from the regression

$$\Delta ExpVol_{i,T} = \beta_0 + \sum_{j=1}^{n} \beta_{1,j} \Delta ExpVol_{i,T-j} + \sum_{j=1}^{n} \beta_{2,j} \Delta RVol_{i,T-j} + \varepsilon_{i,T},$$

where the lag structure is determined by the BIC information criterion. $\Delta ExpVol_{i,T}$ and $\Delta RVOL_{i,T}$ are the stock-level monthly changes in the expected volatility and the realized volatility, respectively. A stock is only included in the analysis if valid stock and options data are available for at least 60 months. There are a total of 3291 unique firms in the sample during the period from 1996 to 2020.

We then explore the return extrapolation effect on expected stock return volatility. If investors have biased volatility expectations at the stock level, then we expect i) the return coefficient ϕ_1 to be significantly negative across all MIDAS regressions with $ExpVol_{i,T}$, VRP_{iT}, $\Delta ExpVol_{i,T,}$ and $\varepsilon_{i,T}$ as the dependent variable, ii) the contribution from the recent past returns to be higher than that from the distant past returns, and iii) the result to be mainly driven by the past negative returns. The MIDAS regression results are reported in Supplementary Material Table IA3.

The results provide strong evidence that the impact of extrapolative beliefs in volatility expectations exists at the individual stock level as well. For the $ExpVol_{i,T}$ regression in Panel A of Table IA3, the median (mean) return coefficient ϕ_1 is -0.158 (-0.159), and 96% of the coefficients are significantly negative at the 5% level. For the VRP_T regression in Panel B, the median (mean) return coefficient ϕ_1 is -0.033 (-0.040), and 90% of the coefficients are significantly negative at the 5% level. For the $\Delta ExpVol_{i,T}$ regression in Panel C, the median (mean) return coefficient ϕ_1 is -0.166 (-0.169), and 97% of the coefficients are significantly negative at the 5% level. For the $\varepsilon_{1,T}$ regression in Panel D, the median (mean) return coefficient ϕ_1 is -0.0158 (-0.161), and 97% of the coefficients are significantly negative at the 5% level.

Moreover, the return extrapolation effects are largely induced by recent past returns. The median total contributions of the recent past 10-day returns are 58%, 65%, 57%, and 57% for the MIDAS regressions from Panels A through D. These contributions are much larger than 45%, as would be the case if the contributions were uniformly distributed over the 22 lagged daily returns. These results are consistent with our findings for the aggregate market returns.

B. The Impact of Past Negative Returns

We next examine whether the asymmetric impact of past negative and positive returns on volatility expectations also exists at the stock level. The results are reported in Table 10. Panels A, B, C, and D report the results for the MIDAS

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TABLE 10

Stock-Level MIDAS Regression: Asymmetric Effects

Table 10 summarizes the results for the firm-level MIDAS regressions. The MIDAS regression is estimated if the firm has valid options and stock data for more than 60-months. In Panel A, the dependent variable is $ExpVol_T$. We follow Bakshi et al (2003) to construct the model-free risk-neutral volatility as a proxy for $ExpVol_T$. In Panel B, the dependent variable is the variance risk premium $VRP_{i,T}$, defined as the difference between variance under risk-neutral expectation and physical expectation. $VRP_T = \mathbb{E}_T^Q(Var_{T,T+1}) - \mathbb{E}_T^P(Var_{T,T+1})$. Following Bollerslev, Tauchen, and Zhou (2009), the realized variance in the month $T-1$ is used as a proxy for $\mathbb{E}_T^P(Var_{T,T+1})$. In Panel C, the dependent variable is the change in volatility expectations, $\Delta ExpVol_{i,T}$, defined as the monthly change in the expected volatility. In Panel D, the dependent variable is the unexpected shocks to volatility expectations. The Unexpected Shocks to Volatility Expectations are the residuals ε_1 from the Vector Autoregression that includes $\Delta ExpVol_T$ and ΔRV_T as the response variables $\Delta ExpVol_T = \beta_0 + \beta_{1,i} \times \Delta ExpVol_{T-i} + \beta_{2,i} \times \Delta RVol_{T-j} + \varepsilon_{1,T}$, where $\Delta ExpVol_T$ is the monthly change of the expected volatility and $\Delta RVOL_T$ is the monthly change of the realized volatility. $Y_{j,T} = \mu_1 + \phi^{Neg} \sum_{t=1}^{22} b_1(t;\theta) Ret_{T-t} D_{T-t}^{Neg} + \phi^{Pos} \sum_{t=1}^{22} b_2(t;\theta) Ret_{T-t} D_{T-t}^{Pos} + \gamma * Y_{j,T-1} + X_{T-1} + \varepsilon_{j,T}$. $D_{T-t}^{Neg}(D_{T-t}^{Pos})$ is a dummy variable that equals 1 if the excess return on day $T-t$ is negative (positive), and 0 otherwise. ϕ^{Neg} (ϕ^{Pos}) captures the relation between the dependent variable and lagged negative (positive) daily excess returns. The Beta polynomial is adopted for the weighting function. $\theta^{Neg} = \left[\theta_1^{Neg}, \theta_2^{Neg}\right]$ and $\theta^{Pos} = \left[\theta_1^{Pos}, \theta_2^{Pos}\right]$ determine the contribution of each lagged daily positive and negative excess returns respectively. The weight of each lagged return is $(t;\theta) = \frac{f\left(\frac{t}{t^{max}}, \theta_1, \theta_2\right)}{\sum_{j=1}^{22} f\left(\frac{t}{t^{max}}, \theta_1, \theta_2\right)}$, with $t = [1, \ldots, 22]$, $t^{max} = 22$. $f(z,a,b) = z^{a-1}(1-z)^{b-1}/\beta(a,b)$, and $\beta(a,b)$ is based on the Gamma function $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. The mean and median negative return coefficient ϕ^{Neg} and positive return coefficient ϕ^{Pos} are reported. The proportion of these return coefficients that are significantly positive, insignificantly positive, significantly negative, and insignificantly negative are reported in columns 3 to column 6. Weights_Lag3 is the cumulative weight from the first 3 lagged daily returns, Weights_Lag5 is the cumulative weight from the first 5 lagged daily returns, and Weights_Lag10 is the cumulative weight from the first 10 lagged daily returns.

	Mean	Median	Significantly Positive %	Insignificantly Positive %	Significantly Negative %	Insignificantly Negative %
Panel A. Risk-Neutral Expected Volatility						
Positive Lagged Ret Coefficient ϕ^{Pos}	−0.068	−0.066	7%	19%	52%	23%
Positive Weights_Lag3	0.462	0.317				
Positive Weights_Lag5	0.578	0.596				
Positive Weights_Lag10	0.769	0.970				
Negative Lagged Ret Coefficient ϕ^{Neg}	−0.206	−0.201	0%	1%	94%	4%
Negative Weights_Lag3	0.220	0.196				
Negative Weights_Lag5	0.354	0.330				
Negative Weights_Lag10	0.660	0.643				
Panel B. Variance Risk Premium						
Positive Lagged Ret Coefficient ϕ^{Pos}	−0.034	−0.031	9%	6%	76%	9%
Positive Weights_Lag3	0.356	0.210				
Positive Weights_Lag5	0.486	0.364				
Positive Weights_Lag10	0.703	0.704				
Negative Lagged Ret Coefficient ϕ^{Neg}	−0.036	−0.028	5%	6%	74%	14%
Negative Weights_Lag3	0.407	0.314				
Negative Weights_Lag5	0.544	0.505				
Negative Weights_Lag10	0.813	0.885				
Panel C. The Change in Volatility Expectations						
Positive Lagged Ret Coefficient ϕ^{Pos}	−0.114	−0.110	3%	8%	75%	14%
Positive Weights_Lag3	0.332	0.183				
Positive Weights_Lag5	0.431	0.309				
Positive Weights_Lag10	0.642	0.611				
Negative Lagged Ret Coefficient ϕ^{Neg}	−0.165	−0.157	0%	1%	92%	7%
Negative Weights_Lag3	0.253	0.212				
Negative Weights_Lag5	0.394	0.363				
Negative Weights_Lag10	0.704	0.700				
Panel D. Unexpected Shocks to Volatility Expectations						
Positive Lagged Ret Coefficient ϕ^{Pos}	−0.130	−0.125	1%	5%	82%	12%
Positive Weights_Lag3	0.281	0.165				
Positive Weights_Lag5	0.380	0.288				
Positive Weights_Lag10	0.607	0.570				
Negative Lagged Ret Coefficient ϕ^{Neg}	−0.143	−0.135	0%	1%	89%	9%
Negative Weights_Lag3	0.272	0.226				
Negative Weights_Lag5	0.414	0.380				
Negative Weights_Lag10	0.721	0.717				

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regression with $ExpVol_{i,T}, VRP_T, \Delta ExpVol_{i,T}$, and ε_T as the dependent variable, respectively. First, the negative return coefficients ϕ^{Neg} are larger (in terms of magnitude) than the positive return coefficients ϕ^{Pos} across all panels. For example, the median coefficients are -0.201 for past negative returns and -0.066 for positive returns in Panel A. Similar patterns can be found in other panels as well. Not only is the magnitude of ϕ^{Neg} larger, but it is also more significant for a larger number of stocks. The proportion of ϕ^{Neg} $\left(\phi^{Pos}\right)$ that is significantly negative at the 5% level is 94% (52%), 74% (76%), 92% (75%), and 89% (82%) across Panels A to D, respectively. Finally, the effects of extrapolating past negative returns are also largely contributed by the most recent returns. The median total weights contributed by the recent past 10-day returns are 64%, 89%, 70%, and 72% across all MIDAS regressions from Panels A to D, respectively.

C. The Impact on Hedging Demand

Similar to the analysis at the index level, we examine the impact of return extrapolation on the hedging demand. The monthly change in the out-of-money put option implied volatility ($\Delta IV_{i,T}^{OTMPut}$) is used as the proxy for investors' hedging demand. We repeat the analysis with $\Delta IV_{i,T}^{OTMPut}$ as the dependent variable in the MIDAS regression. The results are reported in Table IA4. In Panel A, the median (mean) return coefficient ϕ_1 is -0.058 (-0.062), and 73% of the coefficients are significantly negative at the 5% level. Panel B shows that the negative return coefficients, ϕ^{Neg}, have a larger magnitude than the positive return coefficients, ϕ^{Pos}. Collectively, these results suggest that extrapolating lower past returns is associated with higher hedging demand.

Overall, the evidence at the stock level is similar to the market-level finding that investors extrapolate past negative returns when forming expectations about future volatilities.

D. Firm Characteristics

Da, Gurun, and Warachka (2014) show that the investors' extrapolative beliefs about returns are more pronounced amongst firms with larger market capitalization, lower book-to-market ratio, and lower volatility. They also find that return extrapolation is stronger in stocks with better visibility. Given that return extrapolation also affects volatility expectations at the stock level, we now examine whether the impact is more prominent in stocks with the same characteristics as those that drive the return expectations.

At the end of each month T, we estimate the MIDAS regression with a 60-month rolling window and classify a stock into the extrapolative or non-extrapolative group. To be included in the extrapolative group, the following two criteria need to be satisfied: i) the return coefficient ϕ_1 in all MIDAS regressions should be significantly negative, and ii) the relative weight from the recent past 10-day returns in all MIDAS regressions should be above 60%.[25]

[25]The total weight of 45% from the first 10-days past returns is expected if the contributions were uniformly distributed over the 22 lagged daily returns.

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TABLE 11

Stock Characteristics: Extrapolative Versus Non-Extrapolative Groups

Table 11 reports 10 stock characteristics for firms that are classified as the extrapolative and the non-extrapolative group. At the end of each month T, we classify a stock into the extrapolative or non-extrapolative group based on monthly rolling MIDAS regression results. To be included in the extrapolative group, the following three criteria needed to be fulfilled: i) the return coefficient ϕ_1 in $ExpVol_{i,T}$ regression should be significantly negative, ii) the return coefficient ϕ_2 in $VRP_{i,T}$ regression should be significantly negative, iii) the return coefficient ϕ_4 in the $\Delta ExpVol_{i,T}$ regression should be significantly negative, iv) the return coefficient ϕ_3 in the unexpected shocks of volatility expectations $\varepsilon_{i,T}$ regression should be significantly negative, and v) the relative weights from the first 10-lagged returns of these three Midas regressions should be above 60%. The reported stock characteristics include market capitalization (MktCap), book-to-market ratio (BM), idiosyncratic volatilities (Ivol), market beta (Beta), institutional ownership (InstOwn), number of analysts following the firm (Analyst#), the Amihud illiquidity measure (Amihud), percentage bid–ask spread for call options (CallSpread) and put options (PutSpread), and daily return skewness in the month $T-1$ (Skewness). We report the time-series average of the cross-sectional means and medians for the extrapolative group (ExtraGroup) and the non-extrapolative group (NonGroup). The differences and their Newey–West adjusted (18-lags) t-statistics are also reported.

	Mean			Median		
	ExtrapGroup	NonGroup	Diff	ExtrapGroup	NonGroup	Diff
MktCap	1.480	0.846	0.634 (6.95)	1.357	0.753	0.604 (5.86)
BM	0.566	0.582	−0.016 (−2.45)	0.473	0.483	−0.010 (−1.10)
IVol	0.082	0.083	−0.001 (−0.62)	0.072	0.073	−0.001 (−0.35)
Beta	1.045	1.095	−0.050 (−3.84)	0.991	1.039	−0.048 (−2.88)
InstOwn	0.693	0.714	−0.021 (−5.03)	0.720	0.765	−0.045 (−16.43)
Analyst#	11.712	10.736	0.976 (9.28)	10.583	9.217	1.366 (7.95)
Amihud	0.005	0.010	−0.006 (−3.87)	0.001	0.001	0.001 (−5.06)
CallSpread	0.227	0.224	0.003 (0.60)	0.149	0.144	0.004 (1.17)
PutSpread	0.175	0.174	0.000 (0.00)	0.120	0.118	0.001 (0.49)
Skewness	0.063	0.088	−0.025 (−3.92)	0.052	0.073	−0.021 (−4.48)

Once stocks are categorized into the extrapolative and non-extrapolative groups, we compare the following 10 stock characteristics across the two groups, including i) market capitalization (MktCap), ii) book-to-market ratio (BM), iii) idiosyncratic volatility (Ivol), iv) market beta (Beta), v) institutional ownership (InstOwn), vi) number of analysts following the firm (Analyst#), vii) Amihud illiquidity measure (Amihud), viii) percentage bid–ask spread for call options (CallSpread), ix) percentage bid–ask spread for put options (PutSpread), and x) daily return skewness in the month $T-1$ (Skewness). Results are reported in Table 11.

Extrapolative stocks are larger with lower market beta and more analyst coverage. These characteristics point to better visibility and a more transparent information environment. In addition, the extrapolative stocks have lower Amihud illiquidity, indicating better liquidity in the stock markets. Finally, the return extrapolation is stronger in the growth stocks (lower BM ratio). These results present similar characteristics to the extrapolative belief in expected returns documented in Da et al. (2014). The similarity in the firm characteristics echoes the survey results of Amromin and Sharpe (2014) and Kaplanski et al. (2016) that the

extrapolative beliefs about the risk and returns are jointly formed such that investors expect a lower return and higher volatilities following lower returns. Finally, the extrapolative stocks are associated with more negatively skewed stock returns, which is consistent with the prediction of Atmaz (2022).

VI. Conclusion

We provide the first comprehensive empirical evidence that the extrapolation of returns impacts the expectations of future volatility. Investors with extrapolative beliefs tend to overestimate future volatility following lower returns. The volatility expectation bias is larger when the degree of extrapolation is higher. This impact of return extrapolation on volatility is driven by past negative returns and pessimistic sentiment rather than by positive returns or optimistic sentiment. The effect is primarily driven by recent rather than distant past returns, consistent with the distinctive feature of extrapolative beliefs. These findings suggest that it is the pessimistic beliefs of investors following low returns that drive the biased volatility expectations.

The higher volatility expectation following lower past returns leads to a higher variance risk premium, suggesting that investors are willing to pay more to insure against the perceived higher expected volatility. Extrapolating lower past returns results in more positive volatility expectation changes and unexpected volatility expectation shocks, indicating an upward bias in the volatility expectations. At the stock level, the return extrapolation effect on volatility expectations is more pronounced in stocks with better visibility, liquidity, and more negatively skewed returns.

Our findings confirm the survey evidence that investors extrapolate past returns when forming expectations about future volatilities. We also provide support for the model predictions of Atmaz (2022) that biased volatility expectations due to return extrapolation affect option prices. More importantly, our study highlights a new mechanism underlying extrapolative beliefs about expected volatility, that is, biased volatility expectations are driven by investor pessimism, and the bias is larger when the degree of extrapolation is higher.

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Appendix

In this section, we describe the construction of physical volatility expectations measures that include i) VOL_T^{GARCH}, where the month T conditional volatility expectation is forecasted using a GARCH(1,1) model, ii) VOL_T^{DY}, where the month T conditional volatility expectation is forecasted using the predictive regression of Drechsler and Yaron (2011), and iii) VOL_T^{Corsi}, where the the month T conditional volatility expectation is forecasted with the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) of Corsi (2009).

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A. VOL_T^{GARCH}

The conditional volatility expectation VOL_T^{GARCH} is forecasted with a GARCH (1,1) model. The GARCH(1,1) model is written as:

$$ExeRet_t = \mu_t + \varepsilon_t; where\ \varepsilon_t = \sigma_t z_t$$
$$\sigma_t^2 = \beta_0 + \beta_1 \sigma_{t-1}^2 + \beta_2 \varepsilon_{t-1}^2$$

Where $ExeRet_t$ is the daily market excess return. The GARCH(1,1) model is estimated using daily observations over a 60-month rolling window with the last observations at the end of month $T-1$. Once the model parameters are estimated, the multi-period conditional forecast over the month T $\left[\hat{\sigma}_{t+1}^2, \ldots, \hat{\sigma}_{t+n}^2\right]$ are generated. The conditional forecasted annualized volatility $VOL_T^{GARCH} = \left[252 \frac{\sum_{i=1}^{n} \hat{\sigma}_{t+1}^2}{n}\right]^{1/2}$.

B. VOL_T^{DY}

Drechsler and Yaron (2011) show that the future realized variance can be forecasted with a parsimonious model, $RV_T^2 = \beta_0 + \beta_1 RV_{T-1}^2 + \beta_2 VIX_{T-1}^2 + \varepsilon_T$. In the model, RV_T^2 is the realized variance in month T, constructed with 5-minute intraday returns and VIX_{T-1}^2 is the square of the VIX index (divided by 12 to make it a monthly measure) observed at the month end $T-1$. The model is estimated using the monthly observations over a 60-month rolling window with the last observations at the end of month $T-1$. Once the model parameters are estimated, the conditional forecast $\widehat{RV}_T^2$ can be estimated with observations RV_{T-1}^2 and VIX_{T-1}^2. Then $VOL_T^{DY} = \left[\widehat{RV}_T^2\right]^{1/2}$. Drechsler and Yaron (2011) demonstrate that this predictive model generates a superior conditional volatility estimate (VOL_T^{DY}) than the GARCH(1,1) model in fitting future realized variance. Hence, they also use their measure of VOL_T^{DY} to construct the variance risk premium as $VRP_T^{DY} = VIX_T^2 - \left[VOL_T^{DY}\right]^2$ to study the role of long-run risk, rare disasters, and jump risk in determining the variance risk premium.

C. VOL_T^{Corsi}

Corsi (2009) proposes a Heterogeneous Autoregressive Realized Volatility (HAR-RV) model, where the conditional volatility is a linear combination of its autoregressive components in the past daily, weekly, and monthly realized volatility.

$$RV_{t+k}^2 = \beta_0 + \beta_1 RV_t^2 + \beta_2 \sum_{j=0}^{4} RV_{t-j}^2 + \beta_3 \sum_{j=0}^{21} RV_{t-j}^2 + \varepsilon_{t+k},$$

where RV_{t+k}^2 is the k-period ahead realized variance, RV_t^2 is the realized variance on day t, $\sum_{j=0}^{4} RV_{t-j}^2$ is the realized variance in the past week, and $\sum_{j=0}^{21} RV_{t-j}^2$ is the realized variance in the past month. For $k = [1,\ldots,22]$, a k-period a head forecasting model $\widehat{RV}_{t+k}^2$ is estimated using the daily observations over a 60-month rolling window with the last observations at the end of month $T-1$. Once the model parameters (for all $k = [1,\ldots,22]$) are estimated, the multi-period conditional forecast over the month T $\left[\widehat{RV}_{t+1}^2, \ldots, \widehat{RV}_{t+22}^2\right]$ are generated. The conditional forecasted annualized

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volatility is $VOL_T^{Corsi} = \left[252 \frac{\sum_{k=1}^{22} \widehat{RV}_{t+k}^2}{22}\right]^{1/2}$.

With its simple implementation and its performance in capturing short-term variation and long-term memory features in the volatility process, the HAR-RV model of Corsi (2009) has been adopted in several recent papers. For example, Pyun (2019) constructs variance risk premium using VOL_T^{Corsi} to study the return predictability of the variance risk premium. Cheng (2019) examines the pricing of the VIX premium embedded in VIX futures, for which he uses VOL_T^{Corsi}as a measure of the physical expectation.

Supplementary Material

To view supplementary material for this article, please visit http://doi.org/10.1017/S0022109025000249.

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